

A Phase-Sensing Mach–Zehnder Interferometer for Coupling-Regime Characterization of SiV Nanophotonic Cavities

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Abstract

The reflected *intensity* of a single-sided optical cavity cannot distinguish an over-coupled device from an under-coupled one: near critical coupling both produce the same dip. The reflected *phase* can. This report documents a fiber Mach–Zehnder interferometer built to measure the frequency-dependent phase and amplitude of the reflection coefficient of silicon-vacancy (SiV) nanophotonic cavities, together with the electronics, the interleaved lock–probe measurement protocol, and the automated analysis pipeline that turn interference fringes into cavity parameters ($f_{\text{res}}, \kappa_i, \kappa_{\text{ex}}$). The instrument locks the interferometer on a fixed-frequency reference laser, probes with a tunable diode laser, and removes the arm-imbalance background through a measured free spectral range of 725.5 GHz (residual optical path imbalance ≈ 0.41 mm). Repeated single-frequency measurements over 400 lock–probe cycles reproduce the probe phase with a circular standard deviation of 3.1° . Applying the instrument to eleven scans of nine distinct cavities yields joint phase–amplitude fits that separate over- from under-coupled devices at a glance, including strongly over-coupled ($\kappa_{\text{ex}}/\kappa_i \approx 19$) and clearly under-coupled ($\kappa_{\text{ex}}/\kappa_i \approx 0.6$) examples.

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1 Introduction and theory

1.1 Nanophotonic cavities and the coupling regime

Diamond nanophotonic cavities containing silicon-vacancy (SiV) centers are the workhorse light-matter interface of our quantum-network experiments: they funnel photons into a single guided mode, enhance the emitter-photon interaction, and thereby enable spin-photon gates, memory-enhanced communication, and photon-mediated entanglement [1, 2, 3]. A cavity is described by its resonance frequency f_{res} and two loss channels: the *external* coupling rate κ_{ex} into the access waveguide, and the *intrinsic* loss rate κ_i (scattering, absorption, leakage into other modes). Their sum is the total linewidth $\kappa = \kappa_i + \kappa_{\text{ex}}$. The ratio $\kappa_{\text{ex}}/\kappa$ decides how useful the device is as a network node: an *over-coupled* cavity ($\kappa_{\text{ex}} > \kappa_i$) returns most intracavity light to the collection channel, whereas an *under-coupled* one ($\kappa_{\text{ex}} < \kappa_i$) loses it internally. Determining the coupling regime of each fabricated device is therefore a routine but critical screening task.

1.2 Reflection coefficient of a single-sided cavity

For a single-sided cavity probed in reflection through its coupling waveguide, input-output theory [4] gives the amplitude reflection coefficient at detuning $\Delta = f - f_{\text{res}}$ (all rates expressed as frequencies),

$$r(\Delta) = 1 - \frac{\kappa_{\text{ex}}}{\kappa/2 + i\Delta} = \frac{(\kappa_i - \kappa_{\text{ex}})/2 + i\Delta}{(\kappa_i + \kappa_{\text{ex}})/2 + i\Delta}. \quad (1)$$

Its magnitude and phase follow directly:

$$|r(\Delta)| = \frac{\sqrt{\Delta^2 + (\kappa_i - \kappa_{\text{ex}})^2/4}}{\sqrt{\Delta^2 + (\kappa_i + \kappa_{\text{ex}})^2/4}}, \quad (2)$$

$$\varphi_c(\Delta) = \arg r(\Delta) = \text{atan2}\left(\frac{\kappa_i + \kappa_{\text{ex}}}{2}, \Delta\right) - \text{atan2}\left(\frac{\kappa_i - \kappa_{\text{ex}}}{2}, \Delta\right). \quad (3)$$

On resonance, Equation (2) reduces to $|r(0)| = |\kappa_i - \kappa_{\text{ex}}|/\kappa$. The dip depth is invariant under the exchange $\kappa_i \leftrightarrow \kappa_{\text{ex}}$: a reflection-intensity spectrum alone therefore *cannot* tell an over-coupled cavity from an under-coupled one, and near critical coupling ($\kappa_i \approx \kappa_{\text{ex}}$) both look like the same near-total dip. The phase breaks the degeneracy [5]. Following Equation (3) across the resonance:

Over-coupled ($\kappa_{\text{ex}} > \kappa_i$): the numerator of Equation (1) encircles the origin; φ_c winds through a full 360° .

Critically coupled ($\kappa_{\text{ex}} = \kappa_i$): r passes through zero on resonance and the phase jumps by 180° .

Under-coupled ($\kappa_{\text{ex}} < \kappa_i$): φ_c makes a bounded excursion of less than 180° and returns to its baseline.

Figure 1 shows Equations (2) and (3) for the three regimes with identical total linewidth: the lower row of intensity dips is nearly indistinguishable between the outer panels, while the phase spectra in the upper row are qualitatively different. Measuring $\varphi_c(f)$ is thus the direct way to read off the coupling regime—and a phase measurement requires an interferometer, since photodetectors only measure intensity.

1.3 Phase readout with a Mach-Zehnder interferometer

We embed the cavity in one arm of a fiber Mach-Zehnder interferometer (MZI). Writing the fields that arrive at the recombining beamsplitter from the cavity (signal) arm and from the delay

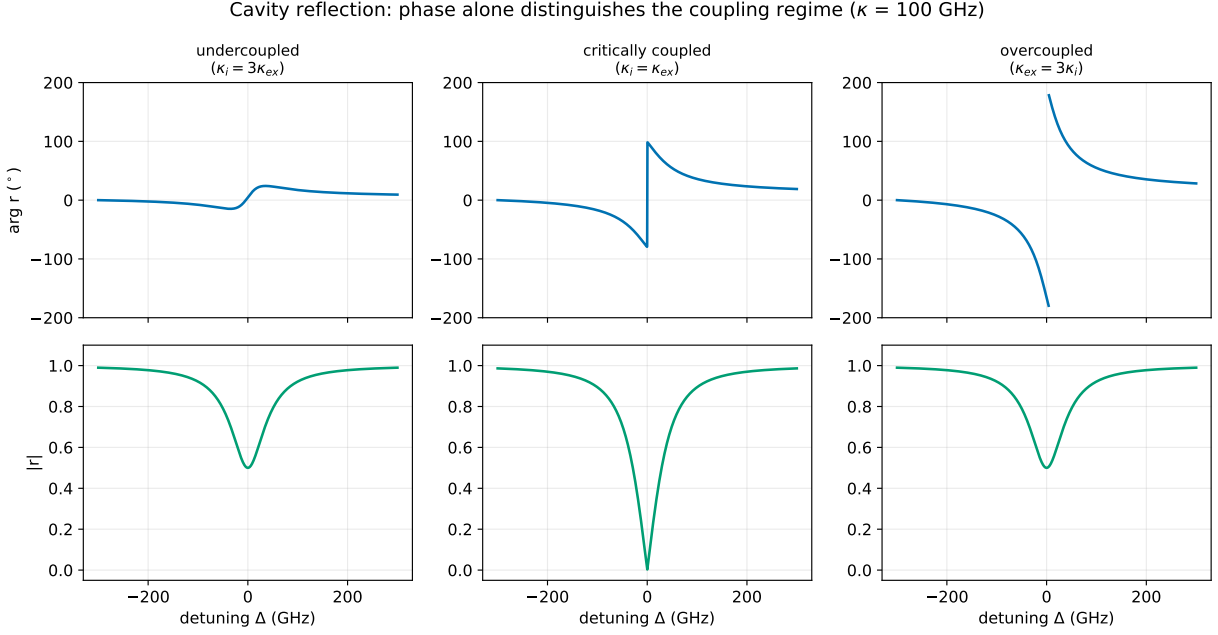


Figure 1: Calculated reflection of a single-sided cavity (Equations (2) and (3), total linewidth $\kappa = 100$ GHz) for an under-coupled ($\kappa_i = 3\kappa_{\text{ex}}$), critically coupled ($\kappa_i = \kappa_{\text{ex}}$), and over-coupled ($\kappa_{\text{ex}} = 3\kappa_i$) device. Top row: reflection phase $\arg r$; bottom row: magnitude $|r|$. The magnitude dips of the two outer columns are identical, while the phase either returns to baseline (under-coupled) or winds through a full turn (over-coupled).

(reference) arm as $E_S \propto a_S r(\Delta) e^{i\varphi_S}$ and $E_L \propto a_L e^{i\varphi_L}$, the two complementary output intensities are

$$I_{1,2}(f) = \bar{I} [1 \pm \mathcal{V}(f) \sin(\Delta\varphi(f))], \quad \bar{I} = \frac{1}{2}(a_S^2 |r|^2 + a_L^2), \quad (4)$$

with fringe visibility

$$\mathcal{V}(f) = \frac{2 a_S a_L |r(f)|}{a_S^2 |r(f)|^2 + a_L^2}, \quad (5)$$

which reaches unity when the two interfering powers are matched, $a_S |r| = a_L$. The total interferometer phase separates into four physically distinct contributions,

$$\Delta\varphi(f) = \underbrace{2\pi f \Delta\tau}_{\text{arm imbalance}} + \underbrace{\varphi_c(f)}_{\text{cavity}} + \underbrace{K v_{\text{mod}}}_{\text{AOM phase control}} + \underbrace{\varphi_0(t)}_{\text{slow drift}}, \quad (6)$$

where $\Delta\tau$ is the residual group-delay difference between the arms, $K v_{\text{mod}}$ is the optical phase written onto one arm by phase-modulating its acousto-optic modulator (AOM) drive with a control voltage v_{mod} (calibration constant K in $^\circ \text{V}^{-1}$), and $\varphi_0(t)$ collects the thermal and acoustic path-length fluctuations of the fibers. The measurement problem is to extract $\varphi_c(f)$ from underneath the other three terms: $K v_{\text{mod}}$ is what we deliberately sweep and servo (Section 3.1), $\varphi_0(t)$ is cancelled by interleaving a phase-locked reference laser (Section 3.1), and the linear background $2\pi f \Delta\tau$ is measured and subtracted through the interferometer free spectral range (Section 3.5). Comparable AOM-based phase control of unbalanced fiber MZIs is standard practice in photonic quantum networking [6, 7, 8].

2 Experimental setup

2.1 Optical layout

Figure 2 shows the optical layout; Figure 4 shows the corresponding hardware. The interferometer proper is fiber-based throughout, with two deliberate free-space excursions: the delay stage and

the cavity microscope.

Laser preparation. Two lasers feed the interferometer from a raised breadboard above the main table (Figure 4b). The *probe* is a Toptica DL pro (“Toptica 2”), located on a neighboring laser table and fiber-coupled to the breadboard through a half-wave-plate/polarizing-beamsplitter chain; a pick-off is routed to a HighFinesse WS7 wavemeter (channel 7), which the control software reads continuously while tuning. The laser is driven by channel 2 of a Toptica DLC pro controller; its motorized grating provides coarse wavelength selection and the piezo provides fine tuning (Section 3.4). The *lock* laser is a Matisse ring laser held at a fixed frequency of 406.545 06 THz (737.415 nm). Each laser passes a manual paddle polarization controller and its own preparation AOM. These two AOMs act purely as fast optical switches: TTL levels from the control electronics select which laser enters the interferometer at any instant (Section 3.1). In-fiber polarizers are unnecessary at this stage because the laser outputs are already well polarized.

Mach–Zehnder interferometer (MZI). The gated beams meet at the input 50:50 fiber coupler BS₁, which both combines the two lasers onto a common mode and splits that mode into the two interferometer arms. Each arm first passes a paddle polarization controller and an in-fiber polarizer, then its arm AOM, which is driven continuously at 200 MHz and serves as the fast phase actuator of the interferometer (Section 2.2).

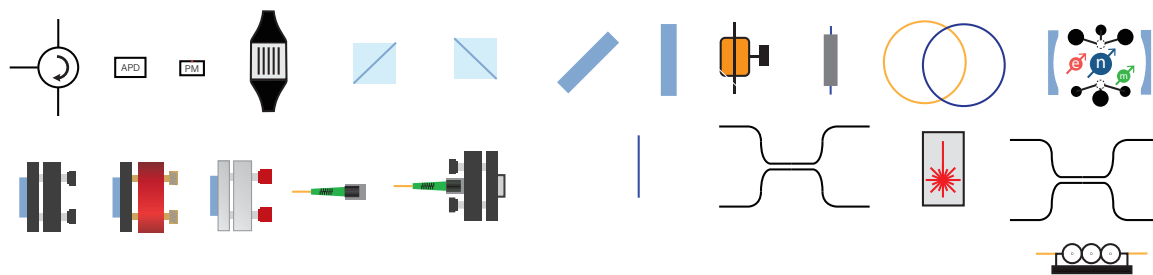
The *delay arm* continues through approximately 8 m of polarization-maintaining patch fiber wound on spools—the coarse path-length match to the longer free arm—and then launches into free space through a collimator and objective. The free-space section (Figure 4c) contains the delay stage: guiding mirrors, a half- and a quarter-wave plate for polarization trimming, and a retroreflector on a 225 mm motorized translation stage. The beam is folded back by the retroreflector, so 1 mm of carriage travel changes the optical path by 2 mm; this is the fine arm-length balance used for free-spectral-range compensation (Section 3.5). The light then couples back into single-mode fiber and proceeds to the recombining coupler.

The *free arm* passes a second in-fiber polarizer and paddle controller and enters port 1 of an optical circulator. Port 2 is spliced to a tapered fiber that couples evanescently to the SiV nanophotonic cavity at the microscope station (Figure 4d): the tapered fiber and the diamond sample sit on independent translation stages, so a device can be selected and touched by the taper under microscope view. The cavity-reflected light returns through the same taper and exits the circulator at port 3—the circulator separates “go” and “return” on a single fiber without the 50% tap loss a passive coupler would impose. After a final paddle controller the reflected light meets the delay arm on the output 50:50 coupler BS₂, whose two complementary outputs are detected by amplified silicon photodiodes PD1 and PD2 (Figure 5b).

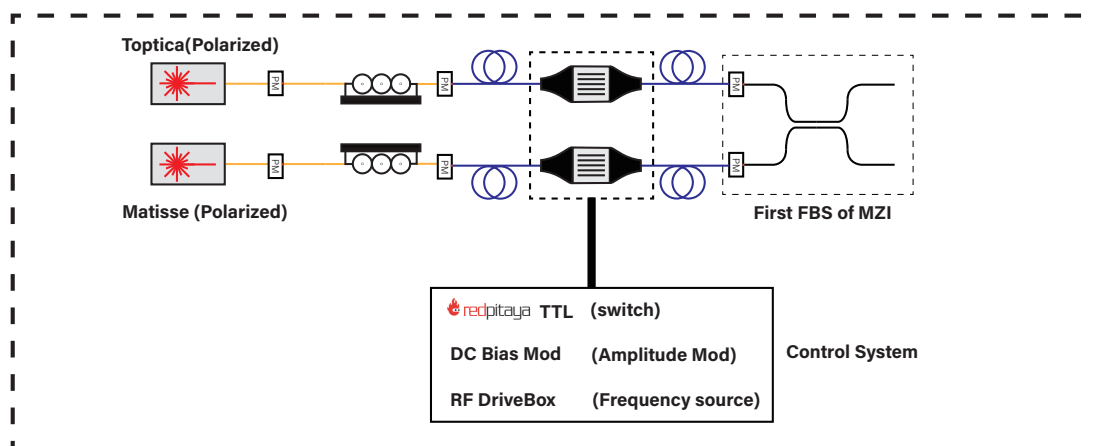
2.2 Electronic layout

A Red Pitaya STEMLab 125-14 orchestrates the entire measurement; it is connected to the control PC over Ethernet, and all bench instruments are additionally under USB or Ethernet remote control by the same PC.

Photodetection and phase actuation. The photodiode outputs are digitized by the Red Pitaya RF inputs IN1 and IN2. The Red Pitaya’s analog output OUT1 is the single phase-control signal of the interferometer: it is amplified by a home-built op-amp stage on a solderless breadboard—an OPA134 with a 4.7 k Ω /1 k Ω feedback network (Figure 6), giving a measured voltage gain of ≈ 6.5 —and fed to the AUX IN connector of a Siglent SDG6032X two-channel function generator. Both SDG6032X channels output 200 MHz sine tones that, after amplification by Mini-Circuits ZHL-1-2W RF amplifiers, drive the free-arm AOM (CH1) and the delay-arm AOM (CH2). CH2 is phase-modulated by the amplified OUT1 signal (external source, 360° deviation at full scale), so the Red Pitaya writes optical phase onto the delay arm with a calibrated



Laser Preparation setup



Toptica 2 Setup

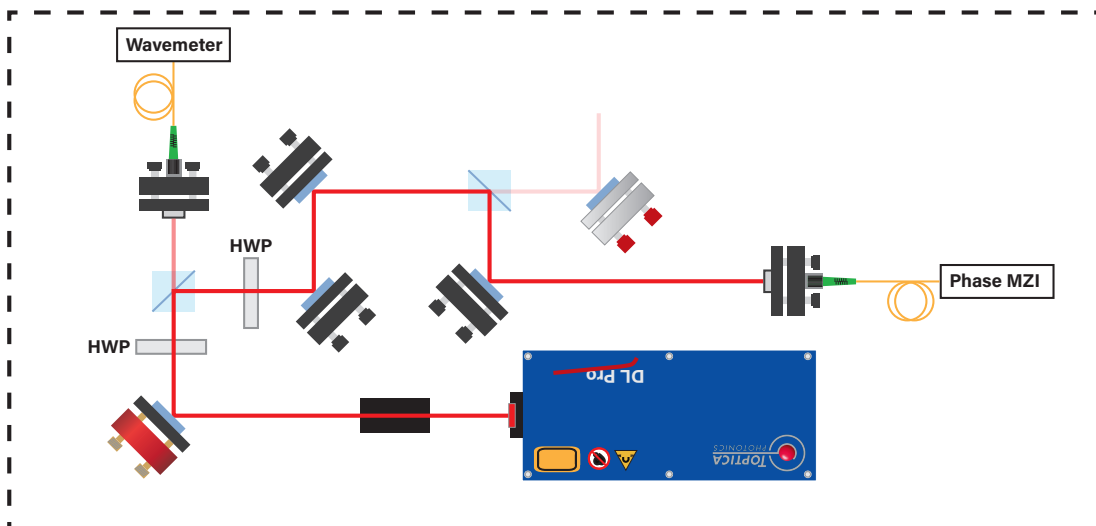


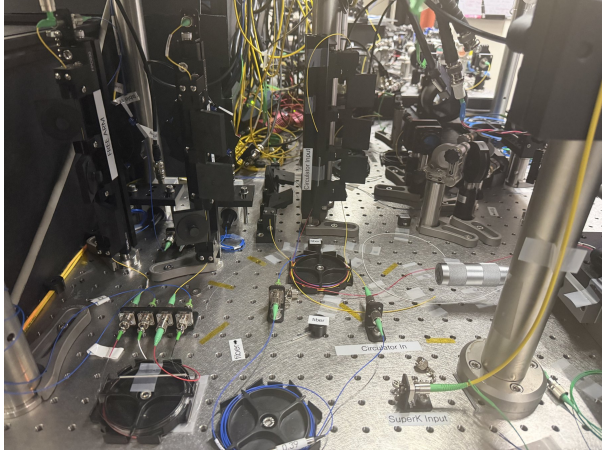
Figure 3: Optical layout of the laser preparation and TOPTICA laser steps.

constant K ($^{\circ}\text{V}^{-1}$); this single knob both closes the lock servo and generates the probe phase sweeps of Section 3.1.

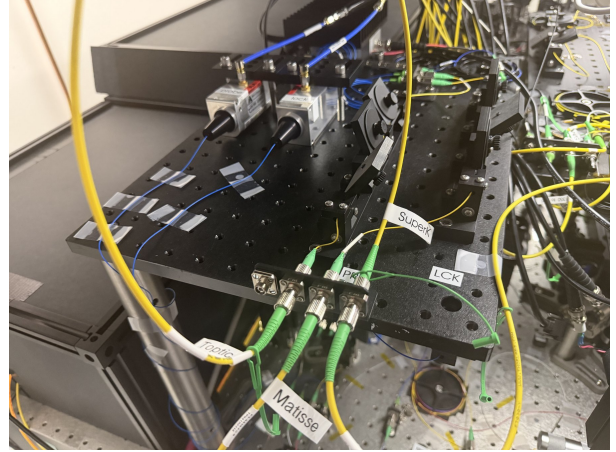
Gating and power. The lock and probe preparation AOMs are driven by fixed-frequency 200 MHz sources (20 dBm) whose TTL inputs are controlled by Red Pitaya digital pins DIO2_P (lock) and DIO4_P (probe)—these two lines implement the lock/probe interleaving. Each fixed source also accepts a DC modulation input, supplied by a Siglent SDG2122X: during calibration the control software trims these DC levels to equalize the optical powers of the two beams (Section 3.2). Two Rigol DP832 supplies power the RF chain: 24 V rails for the ZHL amplifiers and the fixed-frequency drivers, and a ± 12 V dual rail for the op-amp stage. Figure 6 shows the same electronics as a photographic wiring diagram, connection by connection.

2.3 Control software

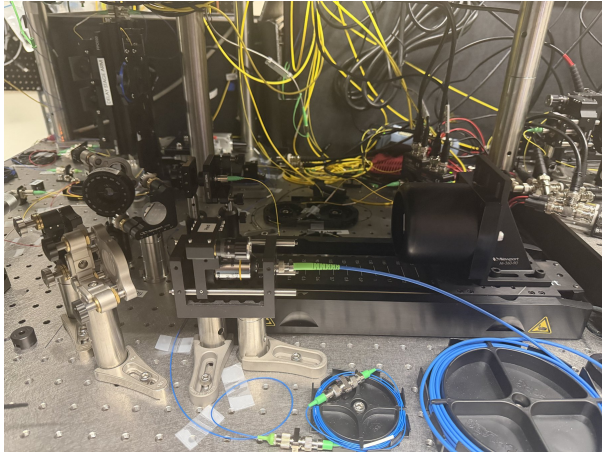
All routines described below—device initialization, system checks, power/lock calibration, the scan engine, FSR measurement, and analysis—run in a dedicated desktop application that automates the measurement end to end and archives every scan with its full hardware configuration. The complete control logic is reproduced as flow diagrams in Section A: probe-laser tuning (Figure 25), the mode check (Figure 26), device connections (Figure 27, the pre-measurement device checks (Figure 28), calibration (Figure 29, the lock–probe measurement cycle and amplitude stages (Figure 30), the FSR measurement (Figure 31), and the analysis pipeline (Figure 32).



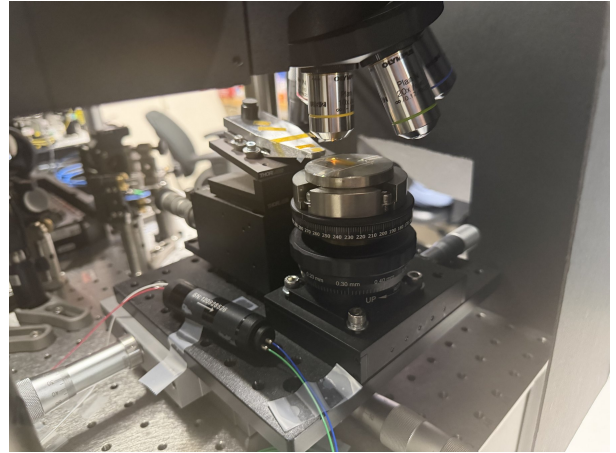
(a) Main MZI board (input coupler and paddle controllers at bottom left, circulator input center).



(b) Laser preparation breadboard: lock and probe gate AOMs with the labeled Matisse / Toptica / lock / probe fiber ports.

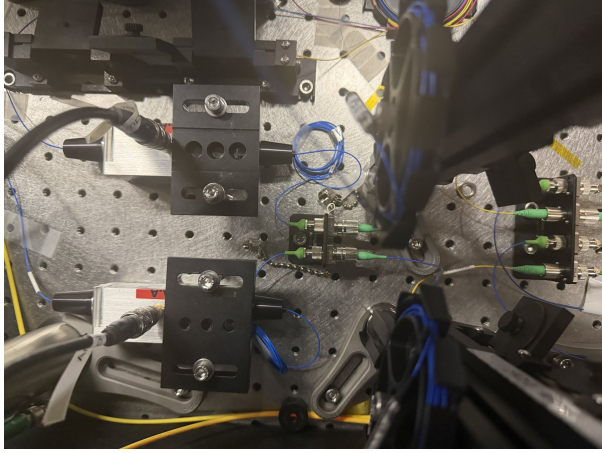


(c) Delay-stage assembly: waveplates, fold mirrors, fiber couplers, the motorized retroreflector rail, and PM fiber spools.

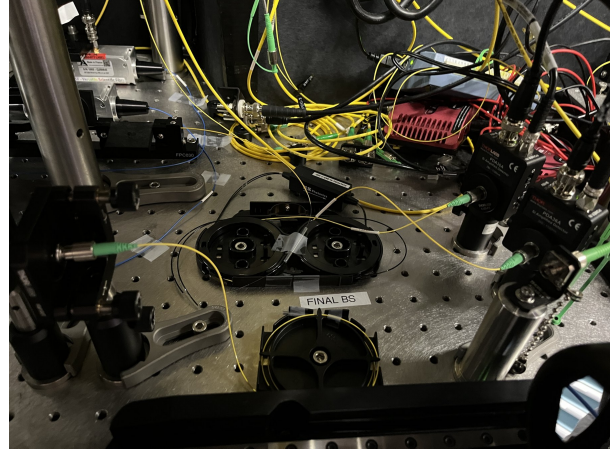


(d) Microscope station: objectives above the diamond sample stage; the tapered-fiber positioner approaches from the left.

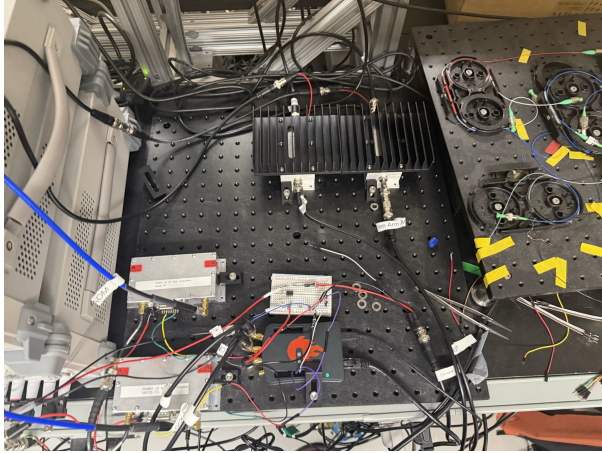
Figure 4: The optical setup.



(a) Arm AOMs and in-fiber polarizers of the two interferometer arms.



(b) Recombining coupler (“final BS”) with the two amplified photodiodes (right).



(c) RF electronics board: Red Pitaya (center), op-amp breadboard, the two ZHL-1-2W amplifiers on heat sinks (top), and the two fixed-frequency AOM drivers (left).



(d) Instrument shelf: two Rigol DP832 supplies (top), Siglent SDG2122X and SDG6032X generators, oscilloscopes, and the control laptop.

Figure 5: Detector, actuator, and electronics hardware.

3 Methods

3.1 Interleaved lock–probe cycle

The fiber paths drift by many optical wavelengths per second (Section 4.1), so the unknown $\varphi_0(t)$ in Equation (6) must be pinned in real time—but it cannot be pinned *with the probe laser itself*: the probe changes frequency at every scan point, carries the cavity phase we are trying to measure, and would make the servo absorb the very signal of interest. We therefore split the two jobs between two lasers and interleave them in time, using the preparation AOMs as fast gates: the fixed-frequency lock laser periodically re-anchors the interferometer, and the probe laser reads the cavity in the short intervals in between. Time interleaving on a single detector pair, rather than running both lasers simultaneously, keeps the two fringe patterns from adding on the photodiodes—they would otherwise have to be separated by RF demodulation or spectral filtering—and guarantees that lock and probe sample the *same* optical path and the same detectors, so slow drifts cancel in the difference (Equation (8) below). One measurement cycle (Figure 7; flow diagram in Figure 30) proceeds in four steps:

1. **Lock window.** Only the lock laser propagates. The Red Pitaya forms the fringe signal $d = \text{in1} - \text{in2}$, which by Equation (4) is $2\bar{I}\mathcal{V}\sin(\Delta\varphi)$: the difference of the two complementary outputs doubles the fringe signal and cancels laser intensity noise, which is common to both photodiodes. A servo closes from d onto OUT1 and holds the interferometer phase at the lock-laser frequency at a fixed setpoint for 0.3 s. The operating point sits mid-fringe, where the response to a phase perturbation is linear and steepest, $\delta d \approx 2\bar{I}\mathcal{V}\delta(\Delta\varphi)$; locking at a fringe extremum would give quadratic (and sign-ambiguous) error signals. The servo is a pure integrator: the disturbances to cancel (thermal, acoustic) live at low frequency, and an integrator provides zero steady-state error there while rolling off gently below the acoustic band; its integral gain is a bench-tuned value carried in the calibration file and rescaled with the measured fringe slope D (Section 3.2), so the realized loop bandwidth survives changes in fringe contrast and optical power.
2. **Freeze and swap.** The servo output is frozen, holding the actuator voltage—and with it the fringe operating point—constant; the TTL gates then extinguish the lock beam and open the probe beam. Freezing, rather than keeping the loop closed, is essential: with the lock beam off there is no valid error signal, and with the probe beam on the servo would try to cancel the cavity phase itself. Over the few milliseconds of a probe window the frozen fringe drifts by only a few degrees (Section 4.1), and even that residual is removed by the correction of step 4.
3. **Probe window.** With the probe laser at f_{probe} , OUT1 executes a triangular sweep around the held value: the swept phase Kv_{asg} carries the fringe through several periods while both photodiodes are recorded. Sweeping the fringe deliberately—instead of reading a single static level—makes the phase estimate immune to slow amplitude and offset drifts: the phase is encoded in the *position* of the fringe pattern along the sweep, not in any absolute voltage. The sweep is cut into its monotonic legs, and each leg of $d(v_{\text{asg}})$ is fitted with

$$d(v) = C + A\sin(Kv + \psi), \quad (7)$$

with the phase-per-volt constant K fixed by calibration, so ψ is the only phase parameter (Figure 8). Fitting rising and falling legs separately and comparing them is a built-in diagnostic: drift during the capture or actuator saturation shows up as a leg-to-leg spread, and the per-leg phases are combined by a circular mean into the cycle phase ψ .

4. **Correction.** Just before the freeze the servo was still holding the lock fringe, so the average lock-window fringe level d_{lock} records where in the fringe the servo actually parked:

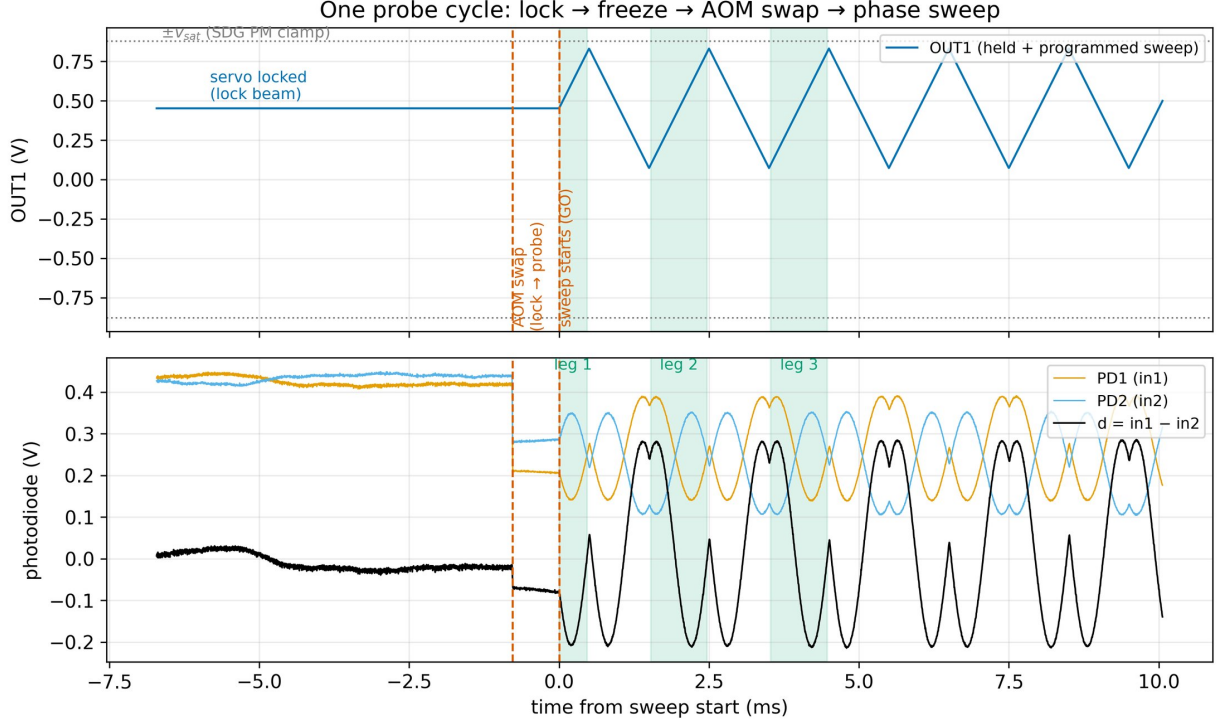


Figure 7: One measured lock–probe cycle. Top: the phase-control voltage OUT1—servo-locked on the lock beam, frozen at the AOM swap, then executing the programmed triangular sweep (the dotted lines mark the $\pm v_{\text{sat}}$ phase-modulation clamp of Figure 10). Bottom: both photodiodes and their difference $d = \text{in1} - \text{in2}$; at the swap the fringe level jumps (the probe beam interferes at a different phase), during the sweep the fringe passes through several full periods, and the shaded legs are fitted with Equation (7).

$\varphi_{\text{lock}} = \arcsin((d_{\text{lock}} - e_0)/a_{\text{lock}})$, with e_0 and a_{lock} from calibration. The cycle’s result is the servo-referenced probe phase

$$\psi_{\text{corr}} = \psi - \varphi_{\text{lock}}, \quad (8)$$

which subtracts the servo’s residual wobble around its setpoint and any slow drift common to both windows. Referencing every probe phase to the servo’s actual parking point, rather than to its nominal setpoint, is what makes consecutive cycles agree at the few-degree level (Section 4.2).

One complete cycle takes roughly half a second, dominated by the 0.3 s lock-settling window; the phase sweep itself uses a 1 ms ramp half-period, fast enough that the frozen fringe barely moves, yet slow enough for the AOM drive chain to follow. The cycle is repeated $N \sim 10$ times per frequency point (up to several hundred for repeatability studies), and the per-cycle phases are averaged with circular statistics—phases live on a circle, and an arithmetic mean across a $\pm 180^\circ$ wrap would be meaningless.

3.2 Calibration

Everything in Section 3.1 rests on knowing, in measured volts, how the actuator moves the fringe and where the fringe sits. Calibration therefore runs immediately before the cycles at every frequency point—not once per day—because its outputs drift with alignment, polarization, and optical power, and because the fringe amplitudes change with $|r(f)|$ as the probe approaches the cavity resonance. A slow triangular ramp on OUT1 (half-period 5 ms, several sweeps averaged) carries the interferometer through more than a full fringe on both photodiodes (Figure 9), and the software extracts from it:

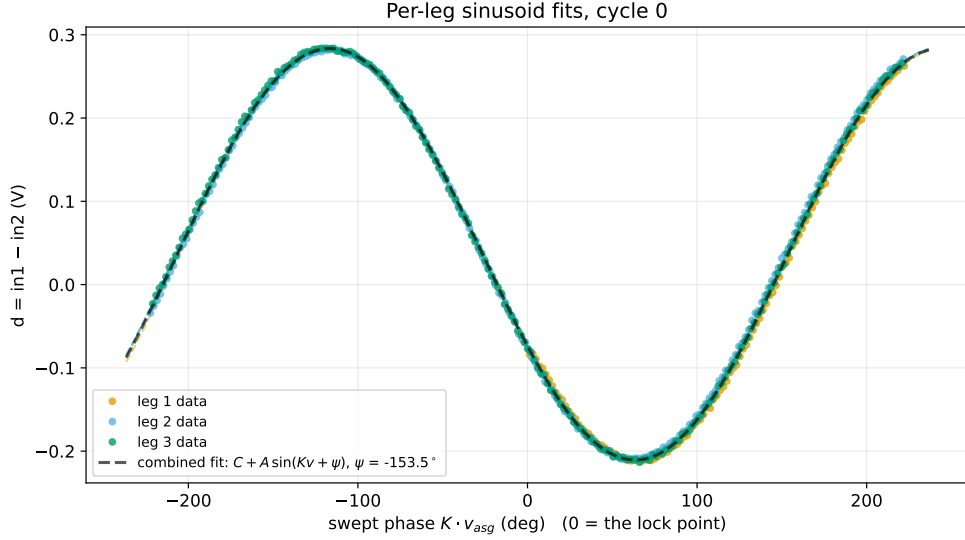


Figure 8: Per-leg sinusoid fits of one probe cycle, plotted against the swept phase Kv_{asg} (zero at the lock point). The three legs collapse onto a single sinusoid, and the combined fit returns the cycle phase ψ .

- the fringe extrema and midpoints $\text{mid}_{1,2}$ of each photodiode, and from them the offset e_0 and amplitude a_{lock} of the difference signal d —the quantities that convert a measured lock level d_{lock} into the parking phase φ_{lock} of Equation (8). (e_0 is nonzero whenever the two fringe midpoints differ, e.g. through an imperfect 50:50 splitting ratio of the output coupler.)
- the phase-per-volt constant K ($^\circ \text{V}^{-1}$) of the actuator chain (OUT1 $\rightarrow$ op-amp $\rightarrow$ SDG phase modulation $\rightarrow$ optical phase), obtained by fitting the swept fringe with K free. Fixing K in all later leg fits (Equation (7)) removes a degeneracy: with both K and ψ free, a short leg cannot distinguish a phase shift from a scale error.
- the fringe slope D (volts of d per volt of OUT1) at the operating point, which converts the servo error into actuator units; the integrator gain is scaled by $1/D$ so the closed-loop bandwidth stays the same whatever the day’s fringe contrast and optical power.
- the fringe visibility of each photodiode, $\mathcal{V}_j = (\max_j - \min_j)/(\max_j + \min_j)$, as a health check of polarization overlap and power balance (Equation (5)).

Servo gains: suggested versus tuned. The controller expresses the integral gain as the integrator’s unity-gain frequency i (in Hz); together with the plant gain D the closed loop crosses unity at $f_{\text{UGF}} \approx i \cdot D$. Each calibration therefore computes a *suggested* integral gain, $i_{\text{sugg}} = f_{\text{UGF}}^{\text{target}}/D$ with a configured target of 200 Hz, purely as a reference point scaled to that day’s fringe. The gains the lock actually runs are the *tuned* pair $(P_{\text{tuned}}, i_{\text{tuned}})$ carried in the calibration file, found empirically during the original bench commissioning and preferred over the suggestion whenever present; the 200 Hz target does not inflate the running loop, whose realized bandwidth with the tuned gain is far below it (of order ten hertz with the current calibration values). The tuned proportional gain is zero, i.e. the loop is deliberately a pure integrator: integral action alone cancels the slow thermal drift with zero steady-state error, while a proportional term would extend loop gain into the acoustic band and, in our commissioning tests, increased rather than decreased the cycle-to-cycle phase scatter—sessions that accidentally ran the proportional fallback showed visibly larger spread, which is why the tuned configuration pins $P = 0$. To keep the tuned gains meaningful as conditions change, the calibration rescales them by $D_{\text{old}}/D_{\text{new}}$ whenever the plant gain shifts by more than 20% between runs, preserving the realized loop bandwidth through changes in fringe amplitude and optical power.

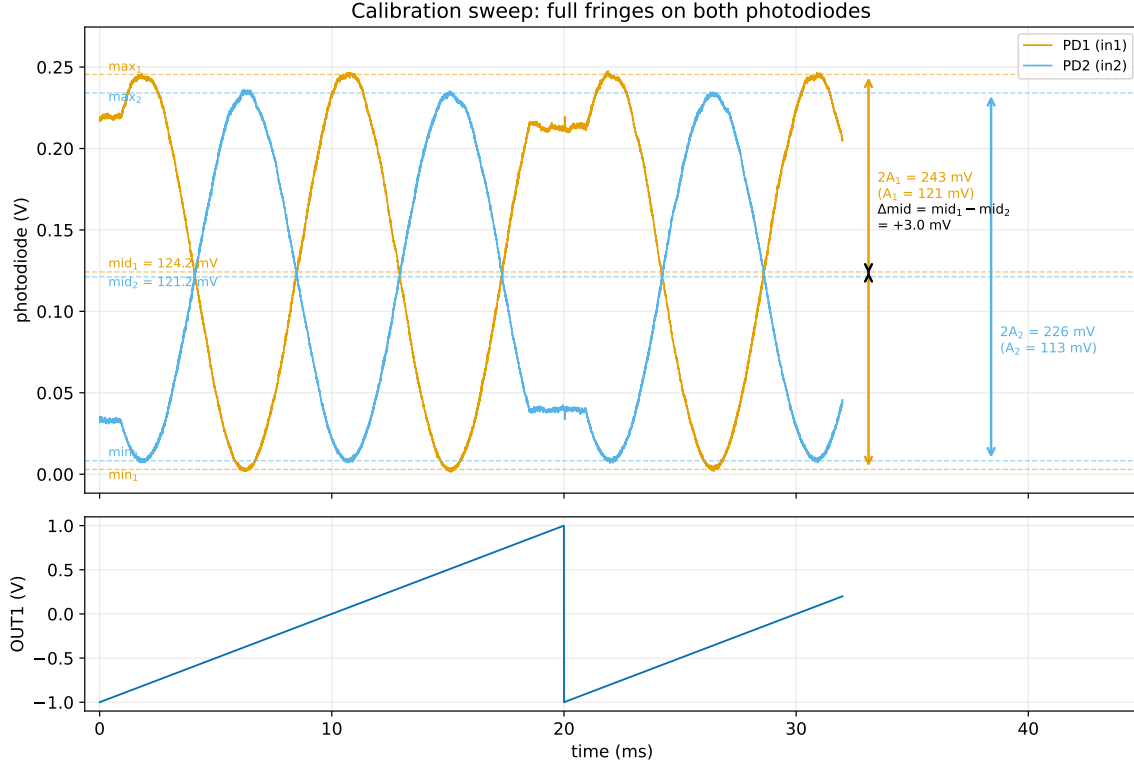


Figure 9: Calibration sweep: a slow OUT1 ramp (bottom) carries both photodiodes through full fringes (top), yielding the fringe extrema, midpoints, amplitudes $A_{1,2}$, and the midpoint offset Δmid annotated on the right.

Saturation of the phase actuator. The SDG6032X phase-modulation input accepts a bounded voltage: beyond $\pm v_{\text{sat}}$ (the amplified voltage that corresponds to the full-scale deviation of $\pm 360^\circ$, i.e. $\pm 360^\circ/K$ at OUT1) the written phase clamps and the fringe freezes. This failure is insidious because nothing else looks wrong: the sweep voltage keeps ramping, the photodiodes keep reporting a plausible level, and only the *fitted* phase is silently biased by the flat-topped fringe. Figure 10 demonstrates the effect deliberately. Calibration therefore measures v_{sat} once per session, and the sweep generator centers the triangular excursion on the held servo voltage and sizes its amplitude to cover a full 2π of phase while staying inside $\pm v_{\text{sat}}$; samples that nevertheless land in the clamped region (e.g. when the servo parks close to a rail) are discarded before the leg fits.

Power equalization and visibility. By Equation (5), unit visibility requires matched interfering powers, while the photodiode inputs must simultaneously stay inside the Red Pitaya’s 1 V range—two goals that pull in opposite directions, since more power buys shot-noise margin but risks clipping. A calibration routine trims the SDG2122X DC levels (preparation AOMs) and the arm-AOM drive amplitudes in two stages: first maximize the optical power of each beam subject to a 0.9 V cap on the *instantaneous* per-photodiode peak, then trim the brighter beam until the two mean powers agree to a ratio of at least 0.8. The peak cap is deliberately measured with both arms interfering rather than from the single-arm powers: on a fringe maximum the fields add coherently, so the instantaneous peak exceeds the sum of the single-arm levels, and a cap based on the latter would still clip. Clipping matters doubly here, because a clipped fringe is no longer sinusoidal and biases every fit built on Equation (7). A separate check records 10 s of free-running fringe per beam (Figure 11) and warns when the visibility falls below 50%, which in practice indicates polarization walk-off between the arms; the paddle controllers are then readjusted before the scan proceeds.

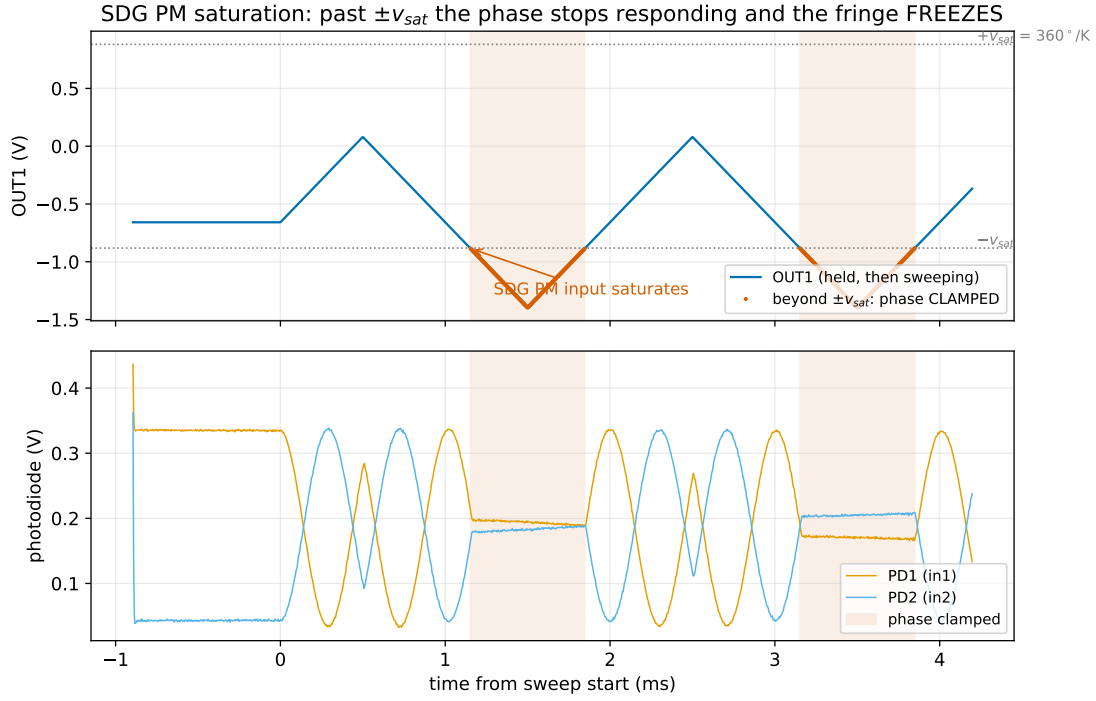


Figure 10: Phase-modulation saturation. When the swept OUT1 exceeds the SDG's $\pm v_{sat}$ input range (top, shaded), the optical phase stops responding and both photodiodes freeze mid-fringe (bottom, shaded)—a failure mode the sweep sizing must avoid.

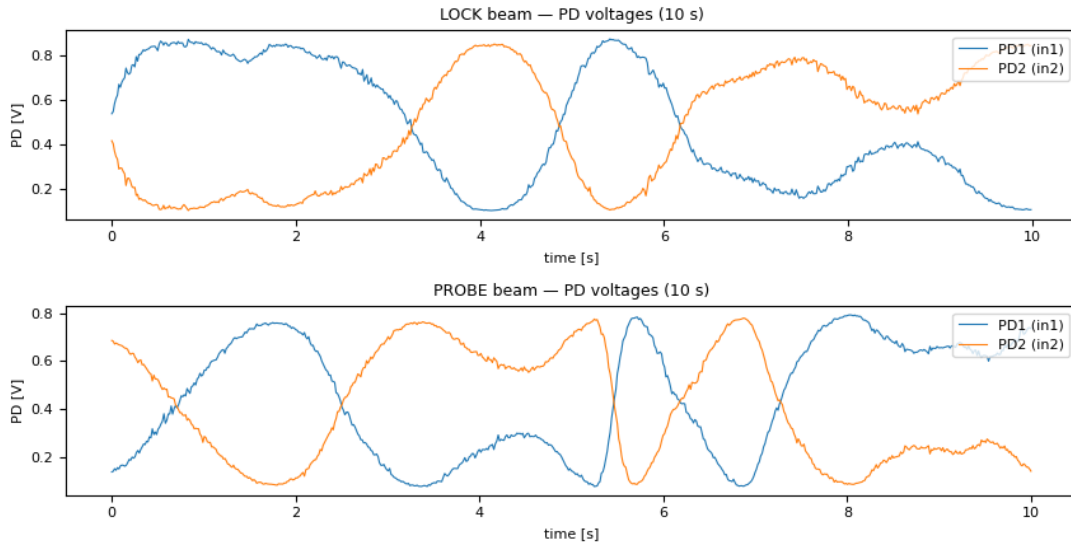


Figure 11: Ten-second free-running fringe records used by the visibility check, for the lock beam (top) and probe beam (bottom). The envelope of each trace gives the fringe visibility of that beam.

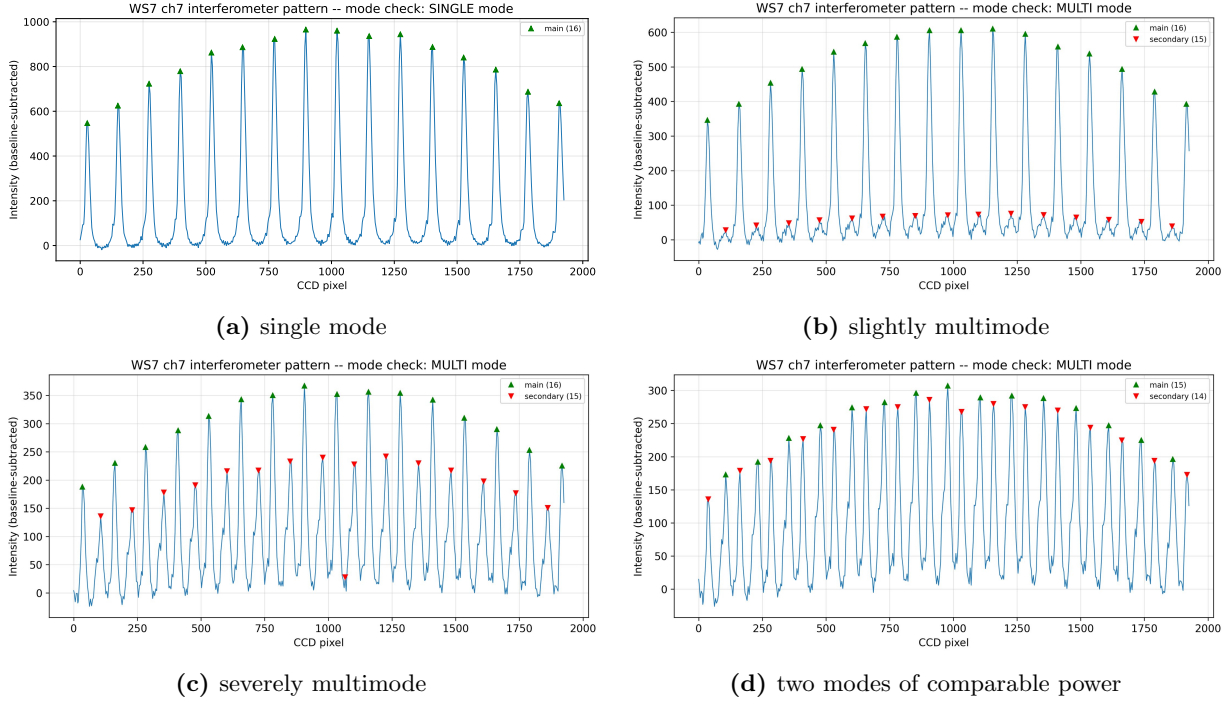


Figure 12: WS7 CCD interferometer patterns behind the mode check, spanning the cases the classifier must separate. (a) A clean single-mode comb: 16 main peaks, no structure between them. (b) Slight multimoding: a weak secondary comb (red markers) appears between the main peaks. (c) Severe multimoding: the secondary comb approaches the main comb in height. (d) The hardest case, two modes of nearly equal power—the main/secondary distinction becomes arbitrary, and the uneven-height and uneven-spacing symptoms carry the verdict.

3.3 Probe-laser mode check

A grating-stabilized diode laser can run multimode near mode-hop boundaries, in which case the wavemeter reading is untrustworthy and the probe phase is meaningless. Before each frequency point the software inspects the WS7’s raw CCD interferometer pattern (Figure 12): peaks are located, the main comb is identified, and four symptoms are scored—secondary-peak height relative to the main comb, uneven main-peak heights, uneven main-peak spacing, and FFT sidebands next to the comb period. A single symptom above threshold flags the point as multimode; the tuner then applies small bias-current steps until the pattern is clean (Figure 26).

3.4 Probe-laser frequency tuning

Scans step the probe laser in 2 GHz increments over hundreds of GHz, far beyond the piezo range, so the tuner coordinates three actuators with very different character: the grating motor (coarse, hysteretic, nm-scale), the piezo (fine, $\pm\sim 10$ GHz), and the diode current (mode trimming). The measured piezo transfer function (Figure 13) shows a smooth static nonlinearity common to up- and down-sweeps plus an open-loop hysteresis of up to ~ 0.5 GHz between sweep directions—enough to matter at our 0.1 GHz parking tolerance. The tuning algorithm (Figure 25) therefore works iteratively on the wavemeter reading rather than on a static conversion: the motor first brings the frequency within ~ 10 GHz of the target, then proportional piezo steps walk the remaining distance, re-measuring after each step, with mode checks and current nudges whenever the pattern degrades, until the wavemeter sits within tolerance. Figure 14 shows a representative convergence, including a motor overshoot-and-return episode handled automatically.

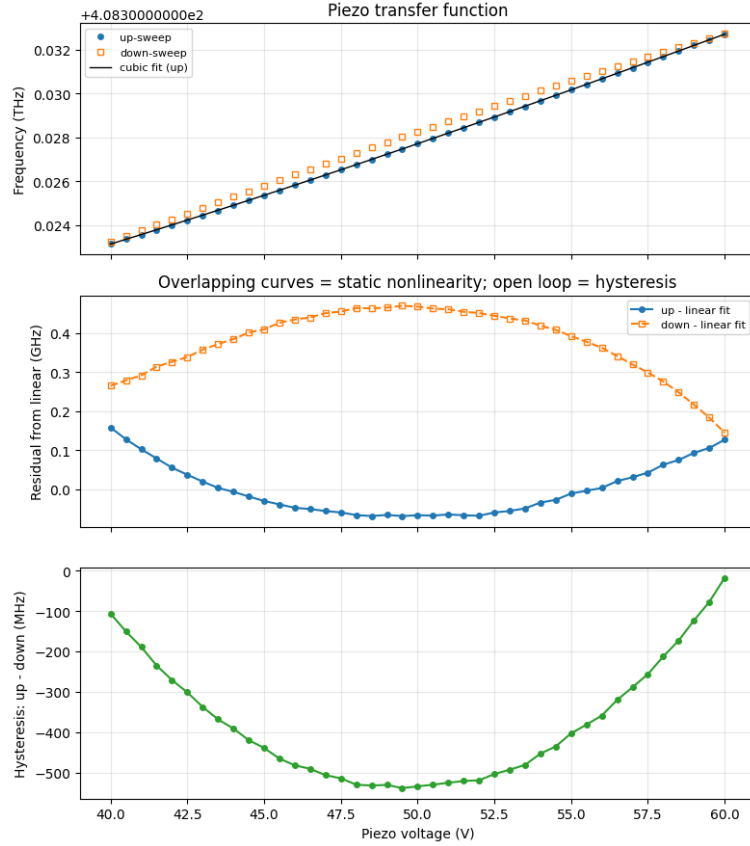


Figure 13: Measured piezo transfer function of the probe laser, up- vs. down-sweep. Top: frequency vs. piezo voltage with a cubic fit to the up-sweep. Middle: residuals from a linear fit—the smooth static nonlinearity is common to both sweep directions. Bottom: the up-down difference, an open-loop hysteresis of up to ~ 500 MHz—the reason tuning iterates on live wavemeter readings instead of trusting a fixed volts-to-frequency map.

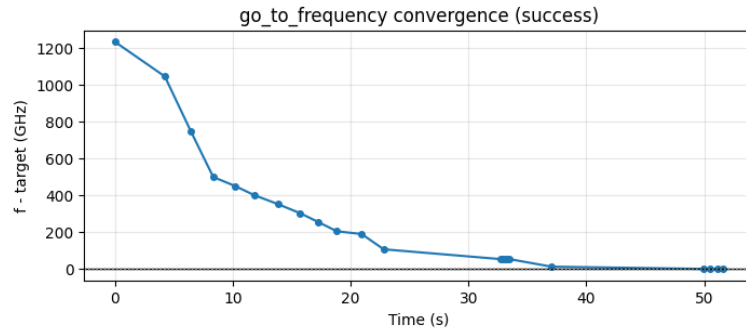


Figure 14: A measured `go_to_frequency` run converging onto a target several THz away: the grating-motor leg closes most of the distance, overshoots, and the piezo walks the remainder into the 0.1 GHz tolerance band.

3.5 Interferometer FSR: bias and compensation

The linear term $2\pi f \Delta\tau$ in Equation (6) makes the measured phase ramp with laser frequency at a rate set by the residual arm imbalance. Expressed per unit frequency, the phase advances by 360° every free spectral range,

$$\nu_{\text{FSR}} = \frac{1}{\Delta\tau} = \frac{c}{\Delta L_{\text{opt}}}, \quad \frac{d(\Delta\varphi)}{df} = \frac{360^\circ}{\nu_{\text{FSR}}}, \quad (9)$$

where ΔL_{opt} is the optical path-length difference (fiber lengths weighted by the group index).

How the background corrupts the data. Figure 15 shows a measured cavity phase profile with a deliberately injected imbalance slope of $3.0^\circ/\text{GHz}$ ($\nu_{\text{FSR}} = 120 \text{ GHz}$): wrapped into the $\pm 180^\circ$ window, the raw data becomes a sawtooth in which the cavity feature is scarcely recognizable. Figure 16 brackets the effect with the same cavity data under a mild ($\nu_{\text{FSR}} = 600 \text{ GHz}$) and a catastrophic ($\nu_{\text{FSR}} = 10 \text{ GHz}$) imbalance. The comparison shows why the slope must be kept small *relative to the scan step*: at 600 GHz the background adds 1.2° per 2 GHz step and the cavity feature survives wrapping essentially intact; at 10 GHz it adds 72° per step, adjacent points are no longer connected by less than 180° , unwrapping becomes ambiguous, and the sampled data is visually indistinguishable from noise. A second, subtler reason to balance the arms well is robustness of the later subtraction: an error $\delta\nu$ in the assumed FSR leaves a residual slope $360^\circ \delta\nu/\nu_{\text{FSR}}^2$ across the scan, which shrinks quadratically with growing ν_{FSR} .

Measuring the FSR. The FSR is measured with the probe tuned far from any cavity resonance, where $\varphi_c(f)$ is flat and Equation (6) reduces to a straight line in f . The spectrogram of Figure 17 shows the raw ingredients of such a run: as the laser steps in 20 GHz increments across a cavity-free 1 THz window, the fringe pattern translates by one full period per ν_{FSR} of laser frequency—the FSR is literally the pitch of the diagonal stripes. The per-step phases of the same run, unwrapped and fitted with a weighted line (Figure 18), give

$$\nu_{\text{FSR}} = \frac{360^\circ}{|\text{slope}|} = 725.5 \text{ GHz}, \quad \Delta L_{\text{opt}} = \frac{c}{\nu_{\text{FSR}}} \approx 0.41 \text{ mm},$$

with a residual scatter of 3.4° rms about the line, consistent with the single-cycle repeatability of Section 4.2 at the ~ 10 -cycle averaging used per step. The two arms of a several-meter fiber interferometer are thus balanced to better than half a millimeter of optical path; at this balance the background ramps at only $\sim 0.5^\circ/\text{GHz}$, so a 2 GHz cavity-scan step advances it by $\sim 1^\circ$ —trivially unwrappable even with 3° -level cycle noise. When a dedicated cavity-free run is not available, the same line can be fitted directly to a cavity scan using only the points away from resonance: Figure 19 shows such a fit on an earlier, less-balanced configuration ($\nu_{\text{FSR}} = 857 \text{ GHz}$), with the cavity's phase excursion visible as the deviation from the wrapped line near the scan center. The full procedure is summarized in Figure 31.

Compensation. Compensation proceeds in two steps. *Physically*, the delay stage is positioned to null most of the imbalance: because the retroreflector is double-passed, the compensating carriage move is half the measured optical mismatch, and the software converts a fitted FSR directly into a stage move. *Digitally*, the small remainder is subtracted in the analysis using the measured ν_{FSR} ; any leftover error is absorbed by the free residual-slope parameter m of the cavity fit (Section 3.6). After a stage move the measurement is repeated once to verify that the fitted slope has collapsed toward zero—the iteration converges immediately because the stage calibration is much better than the required half-millimeter accuracy.

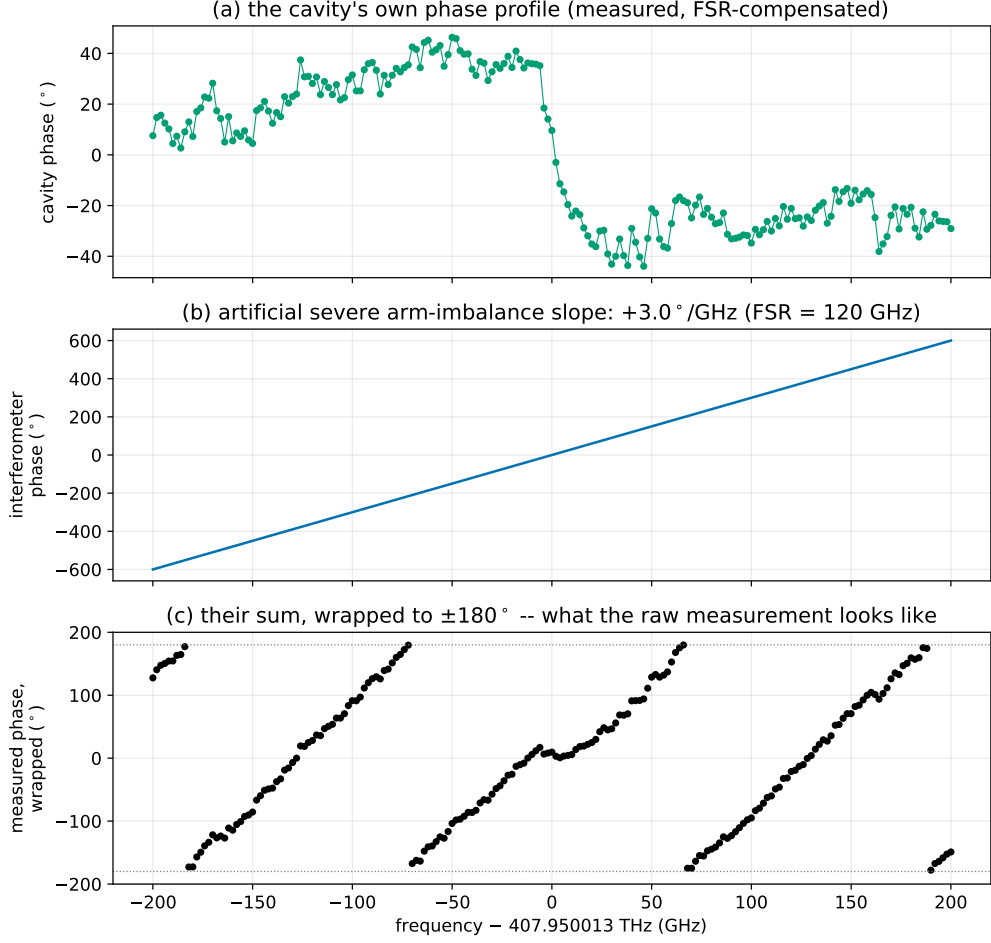


Figure 15: Why the FSR must be compensated. (a) A measured, FSR-compensated cavity phase profile. (b) A deliberately severe arm-imbalance slope of $3.0^\circ/\text{GHz}$ ($\nu_{\text{FSR}} = 120 \text{ GHz}$). (c) Their sum wrapped to $\pm 180^\circ$ —the form raw data takes when the arms are badly unbalanced.

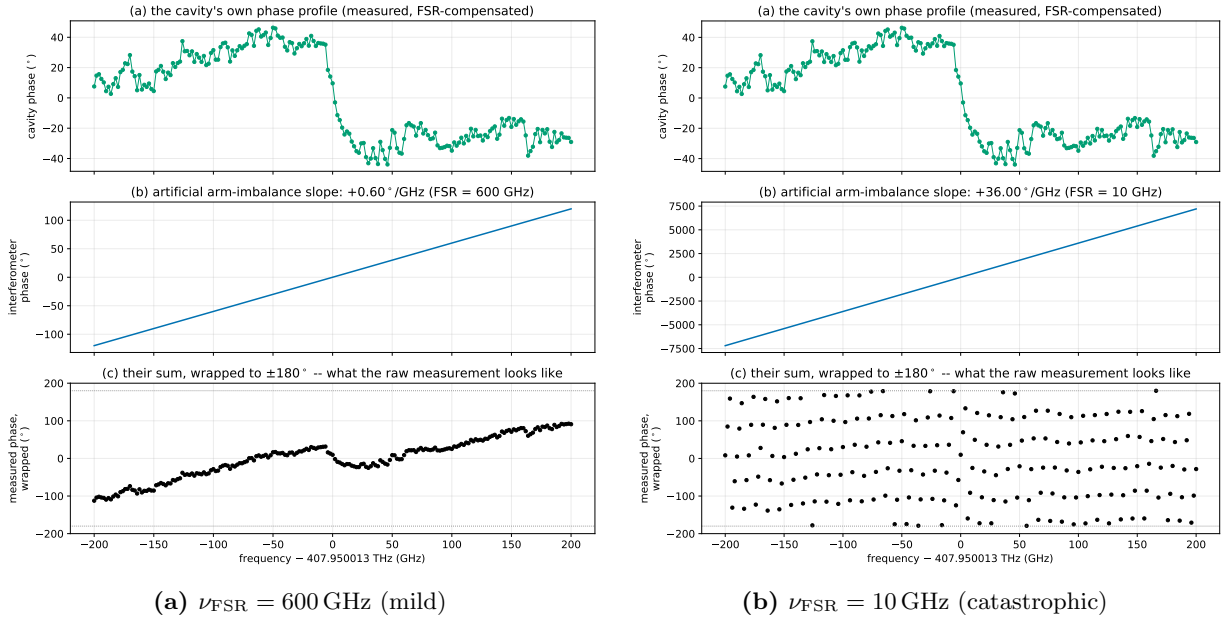


Figure 16: The same cavity data as Figure 15 under (a) a mild and (b) a catastrophic arm imbalance (panels as in Figure 15). At $\nu_{\text{FSR}} = 600 \text{ GHz}$ the wrapped data still shows the cavity feature on a gentle tilt; at 10 GHz the background advances 72° per 2 GHz step and the wrapped samples scatter beyond recognition—unwrapping is no longer possible at this sampling.

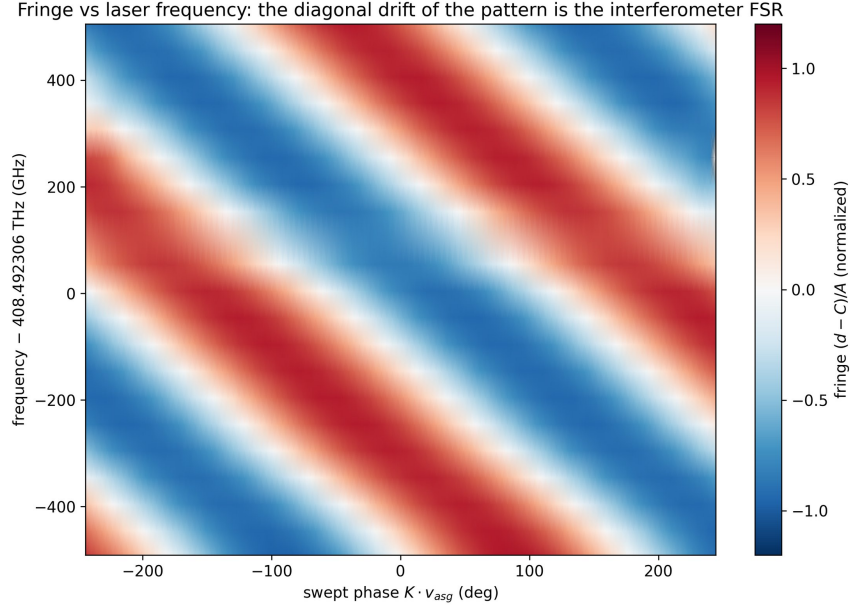


Figure 17: Normalized fringe signal versus swept phase (horizontal) and probe frequency (vertical) over a cavity-free 1 THz range. The diagonal drift of the pattern is the interferometer FSR: one full fringe period of translation per ν_{FSR} of laser frequency.

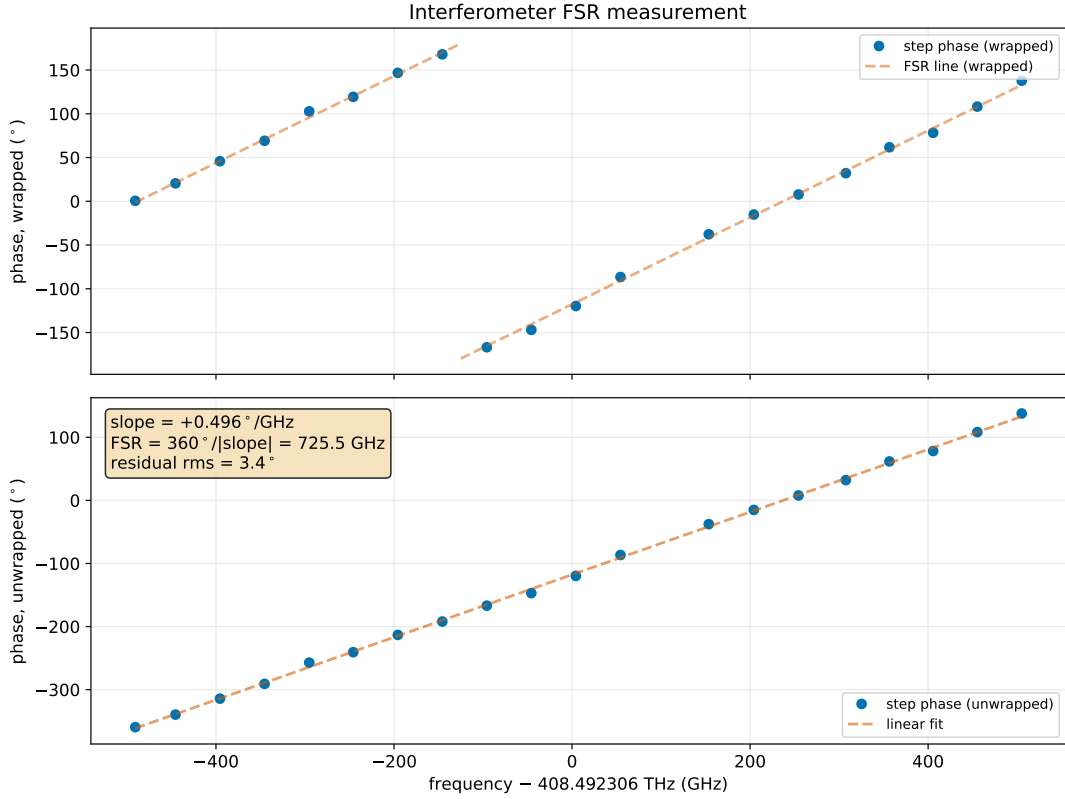


Figure 18: Interferometer FSR measurement on the run of Figure 17, after delay-stage balancing. Top: wrapped per-step phases and the wrapped fitted line. Bottom: the unwrapped phases with the linear fit, giving $\nu_{\text{FSR}} = 725.5$ GHz (slope +0.496 $^{\circ}$ /GHz, residual rms 3.4 $^{\circ}$).

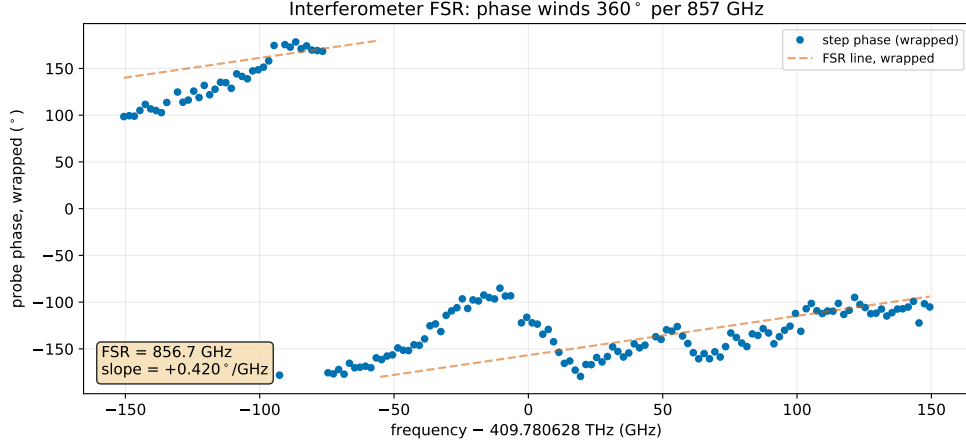


Figure 19: FSR line fitted directly to a cavity scan (wrapped step phases), from an earlier, less-balanced configuration: $\nu_{\text{FSR}} = 856.7 \text{ GHz}$ (slope $+0.420^\circ/\text{GHz}$), fitted on the points away from resonance. The systematic deviation from the dashed line near the scan center is the cavity’s own phase feature riding on the linear background.

3.6 Analysis pipeline and cavity fitting

For each frequency step the recorded cycles are reduced as follows: per-cycle leg fits (Equation (7)) give ψ_{corr} values (Equation (8)); cycles deviating more than 30° from the step’s circular median are discarded as outliers (servo glitches, mode hops); the surviving cycles form the step phase (circular mean) and its uncertainty (circular standard deviation). Separately, an arm-selective amplitude measurement at each step records the mean photodiode powers with only the probe beam, only the lock beam, and neither beam on, giving the normalized cavity amplitude

$$A_{\text{cav}} = \sqrt{\frac{P_{\text{probe}} - P_{\text{dark}}}{P_{\text{lock}} - P_{\text{dark}}}} \propto |r(f)|, \quad (10)$$

in which laser-power drifts common to both beams cancel.

The step phases are unwrapped, the measured FSR line is subtracted, and the phase and amplitude spectra are fitted *jointly* with the eight-parameter model

$$\varphi_{\text{model}}(F) = \arg r(s_{\pm}(F - f_{\text{res}}); \kappa_i, \kappa_{\text{ex}}) + c_0 + mF, \quad (11)$$

$$A_{\text{model}}(F) = s \left(1 + b \frac{F}{\text{span}} \right) \sqrt{(1 - g) |r(F - f_{\text{res}})|^2 + g}, \quad (12)$$

where F is the frequency relative to the scan center, c_0 is an arbitrary global phase offset, m absorbs any residual linear slope left by an imperfect FSR value, s is the amplitude normalization, b a linear tilt of that normalization across the scan (arm coupling and AOM efficiency vary slowly with frequency), and g the incoherent background power fraction—light that never enters the cavity mode and floors the dip at $\sqrt{g}$. The sign $s_{\pm}$ of the detuning axis and the coupling regime are decided by refitting under both hypotheses and keeping the lower χ^2 . The fit returns f_{res} , κ_i , and κ_{ex} with their covariances.

4 Results

4.1 Interferometer stability and lock performance

Figure 20 characterizes the unlocked interferometer. Left to itself, the fringe wanders through several full periods over $\sim 2 \text{ s}$ —a phase drift of order 1000° per second—driven by thermal and acoustic fluctuations of the fiber paths; the amplitude spectral density of the fringe signal shows

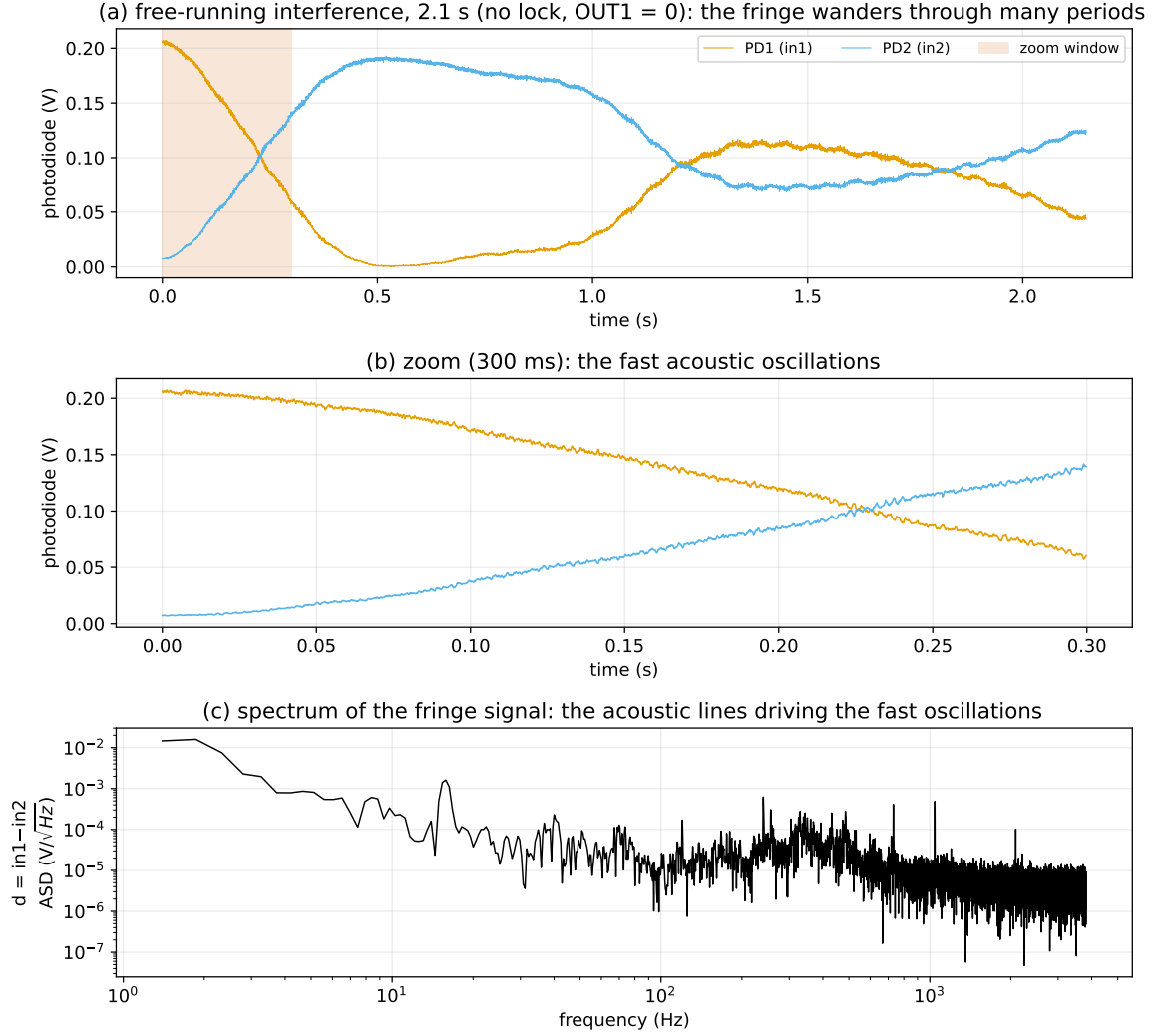


Figure 20: Free-running stability of the interferometer. (a) Both photodiodes over 2.1 s with the servo off: the fringe wanders through many periods. (b) 300 ms zoom resolving the fast acoustic oscillations riding on the slow drift. (c) Amplitude spectral density of the fringe signal d .

the low-frequency drift continuum, discrete acoustic lines (e.g. near 15 Hz), and a broad acoustic band at 100–1000 Hz. This is the environment the lock must suppress and the reason a free-running phase measurement would be meaningless.

Engaging the servo (Figure 21) pulls both photodiodes to the mid-fringe setpoint within ~ 5 ms and holds them there; the actuator settles to a quasi-static value that tracks the slow drift. In lock, the residual fringe fluctuations correspond to a phase jitter far below the free-running excursions, and—more importantly for the measurement—whatever wobble survives is common to the lock and probe windows and is removed by the servo-referencing of Equation (8).

4.2 Phase repeatability

Figure 22 shows 400 consecutive lock–probe cycles at a fixed probe frequency. The servo-referenced phase ψ_{corr} scatters with a circular standard deviation of 3.11° around -110.7° with no visible systematic drift across the ~ 3 -minute record—this is the single-cycle phase resolution of the instrument, and averaging N cycles improves the step uncertainty by $\sim 1/\sqrt{N}$. The same figure overlays the four internal phase estimators (from in1 alone, from in2 alone, from the difference-signal leg fit, and from the frozen fringe at the swap instant): they track each other closely, confirming that the result does not depend on which photodiode combination is used. The

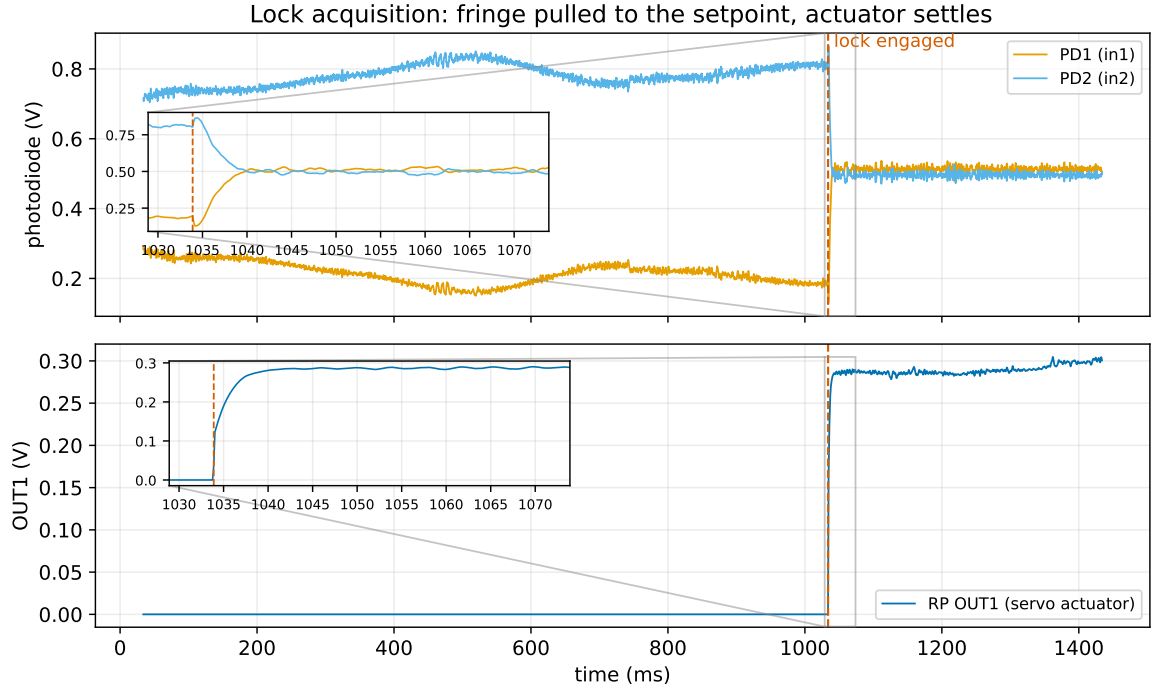


Figure 21: Lock acquisition. At the dashed line the integrator is engaged: both photodiodes (top) are pulled from their free-running wander to the mid-fringe setpoint within a few milliseconds (insets, -5 ms to 40 ms around engagement), and the actuator (bottom) settles to the quasi-static value that cancels the instantaneous path-length drift.

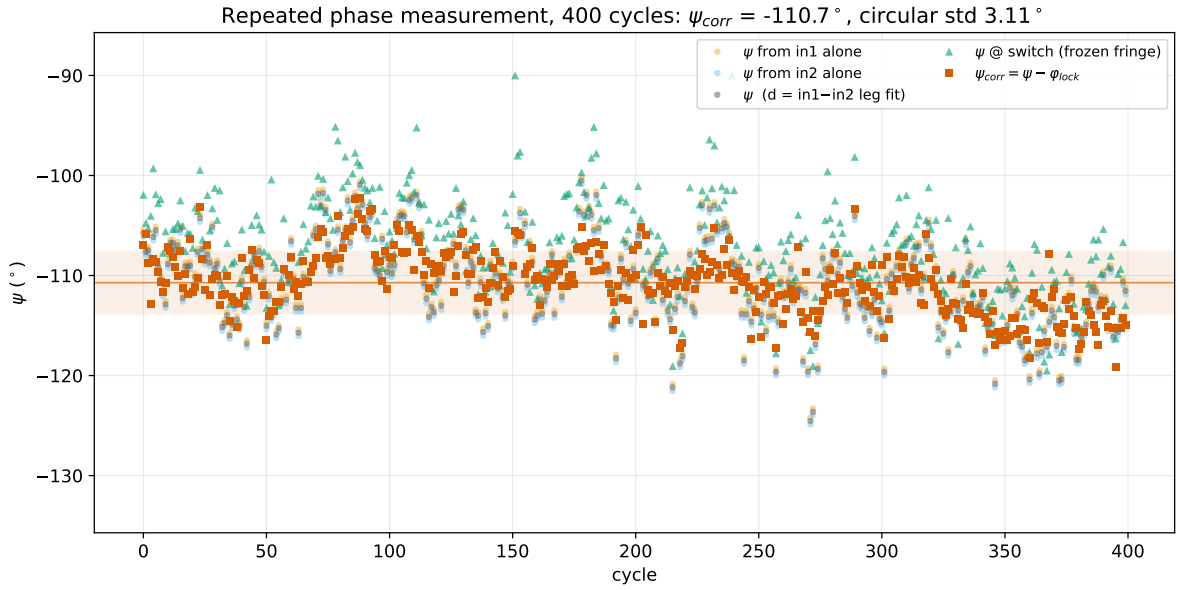


Figure 22: Repeated phase measurement at a fixed probe frequency: 400 cycles, shown for all four internal estimators. The servo-referenced ψ_{corr} (red squares; band = circular mean $\pm$ circular std) reproduces to 3.11° cycle-to-cycle.

uncorrected estimators drift slightly more than ψ_{corr} , quantifying what the servo-referencing removes.

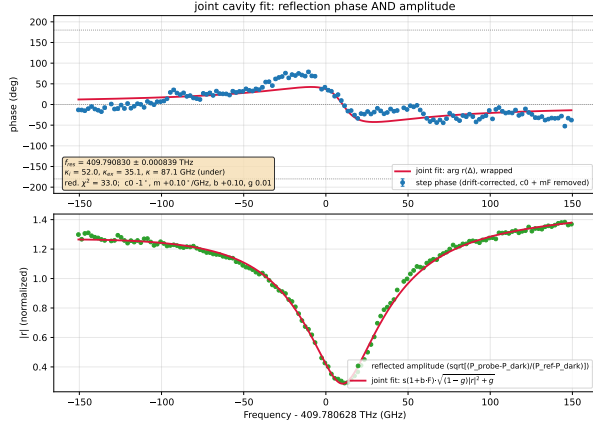
Table 1: Joint phase-amplitude fit results for all analyzed scans. N is the number of frequency steps. Devices near critical coupling ($\kappa_i \approx \kappa_{\text{ex}}$) can flip regime label between runs, as expected. Rows marked [‡] were measured with the desktop application (sessions of August 25–27) and FSR-compensated with the re-measured 777.5 GHz value, as was EG9_LB_b11; all other scans used the earlier 722 GHz value. The free slope m absorbs the difference either way (Section 3.5), so the extracted cavity parameters do not depend on this choice.

Cavity	N	f_{res} (THz)	κ_i (GHz)	κ_{ex} (GHz)	κ (GHz)	regime
EG9_LA_b12	94	411.4958	61.8	38.1	99.9	under
EG9_LA_b14	148	409.7908	52.0	35.1	87.1	under
EG9_LB_a09	150	408.2580	37.5	46.1	83.6	over
EG9_LB_b02	95	410.8393	50.9	34.6	85.5	under
EG9_LB_b04	201	407.9512	50.8	30.8	81.6	under
EG9_LB_b08	146	407.8933	64.8	55.4	120.2	under
EG9_LB_b10	138	408.4997	23.3	27.7	51.0	over
EG9_LB_b11	201	410.9917	52.6	81.2	133.8	over
EG9_LB_b15	122	409.6426	34.2	62.2	96.4	over
EG9_LC_b02	178	409.5129	36.3	40.7	77.0	near-critical
EG9_RB_c03	199	412.0640	47.1	92.8	139.9	over
EG9_RB_c08	150	410.5601	70.5	67.9	138.4	near-critical
EGST3_LA_c01	151	407.7548	4.5	86.6	91.0	over
EGST3_LA_c015	149	411.6418	33.5	63.2	96.7	over

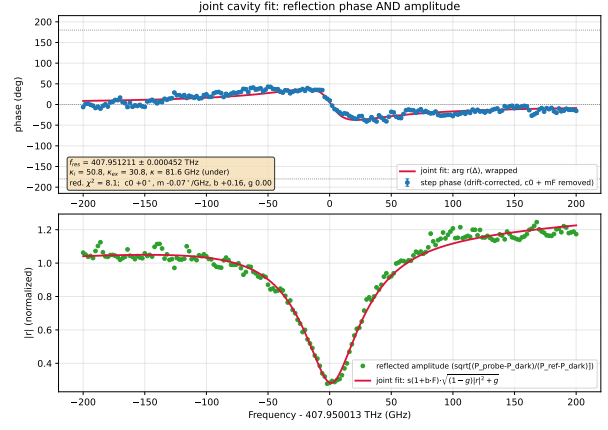
4.3 Cavity phase and amplitude spectra

We applied the full protocol to eleven scans covering nine distinct nanophotonic cavities across three samples. Each scan comprises 146–201 frequency steps of joint phase and amplitude data; all were processed identically (outlier threshold 30° , probe-beam amplitude normalization, manual-FSR subtraction with the free residual slope m of Equation (11) absorbing any small FSR error). Table 1 collects the fitted parameters of all eleven scans, and Figure 23 shows the fits of the six EG9-sample devices.

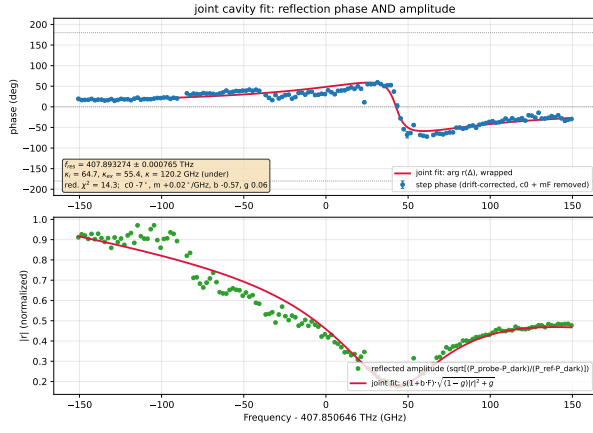
The results span the full range the theory of Section 1 predicts. On the over-coupled side, EG9_RB_c03 and EG9_LB_b11 (Figure 23) wind their phase through nearly a full turn while their amplitude dips stay well above zero, and EGST3_LA_c01 is the most extreme case in the set ($\kappa_{\text{ex}}/\kappa_i \approx 19$, Table 1). EG9_LB_b04 and EG9_LA_b14 are clearly under-coupled: bounded phase excursions that return to baseline. EG9_RB_c08 sits near critical coupling—its two independent measurements give $\kappa_i \approx \kappa_{\text{ex}}$ within 3 GHz and accordingly land on opposite sides of the regime boundary, exactly the ambiguity a magnitude-only measurement would leave everywhere and that the phase confines to the critical point itself. The repeated devices also probe day-to-day reproducibility: for EG9_RB_c08 the two runs agree on κ within $\sim 15\%$ and on f_{res} within 26 GHz; for EG9_LA_b14 the resonance moved by 45 GHz between sessions. We attribute these shifts to changes in the device and its environment between sessions (e.g. the tapered-fiber contact position and slow spectral drift of the structures) rather than to the instrument, whose single-session repeatability is at the few-degree, sub-GHz level (Section 4.2); a systematic study of this drift is beyond the scope of this report.



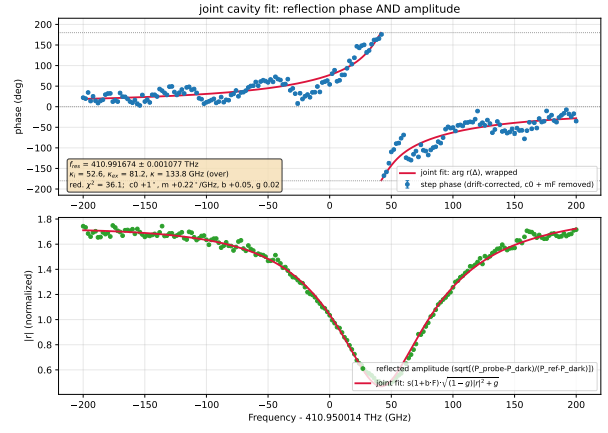
(a) EG9_LA_b14 (under-coupled)



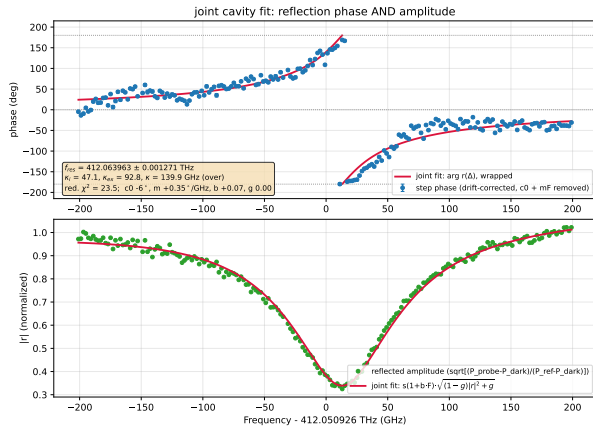
(b) EG9_LB_b04 (under-coupled)



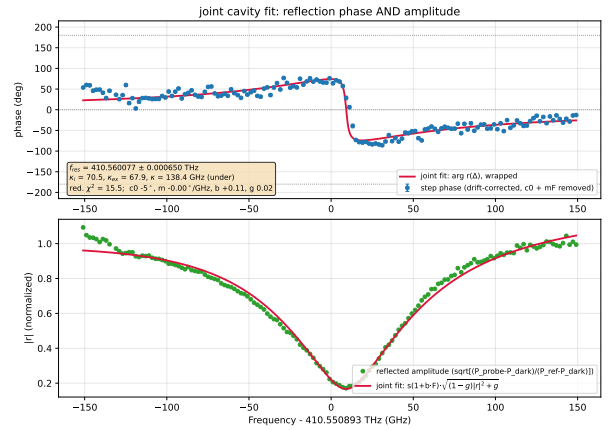
(c) EG9_LB_b08 (under-coupled)



(d) EG9_LB_b11 (over-coupled)

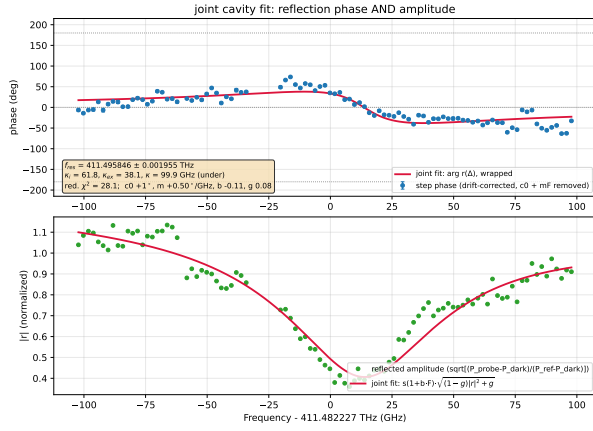


(e) EG9_RB_c03 (over-coupled)

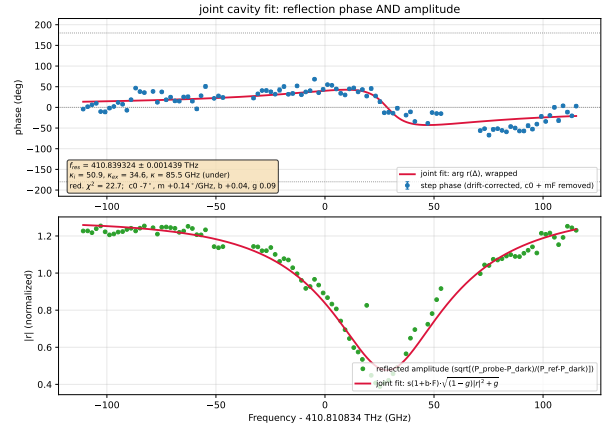


(f) EG9_RB_c08 (near-critical)

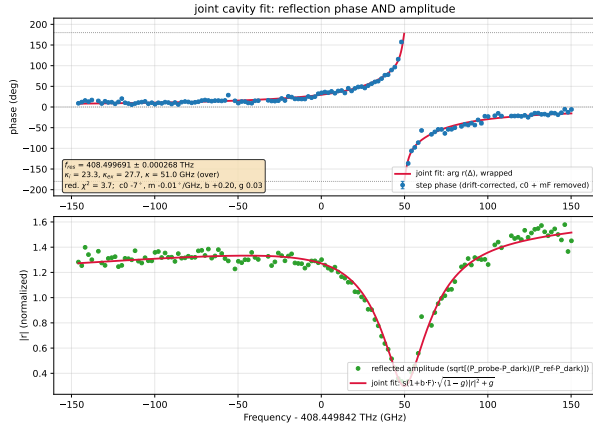
Figure 23: Joint phase–amplitude cavity fits for the six EG9-sample devices (per panel: top, unwrapped FSR-compensated phase with the fitted $\arg r + c_0 + mF$; bottom, normalized amplitude with the fitted Equation (12)). Over-coupled devices wind the phase through a full turn; under-coupled devices make a bounded excursion that returns to baseline. Fit parameters for all scans, including the three devices not shown here, are given in Table 1.



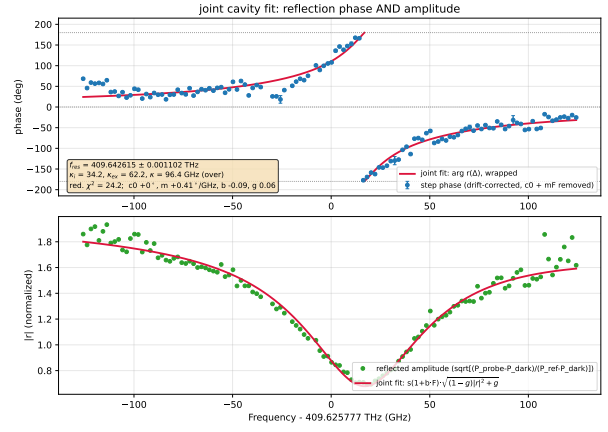
(a) EG9_LA_b12 (under-coupled)



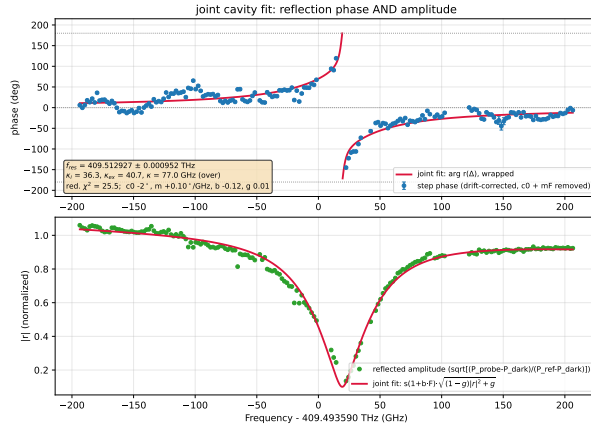
(b) EG9_LB_b02 (under-coupled)



(c) EG9_LB_b10 (over-coupled)



(d) EG9_LB_b15 (over-coupled)



(e) EG9_LC_b02 (near-critical)

Figure 24: Joint phase–amplitude cavity fits for the five EG9 devices measured with the desktop application in the sessions of August 25–27 (same panel layout as Figure 23). These scans were FSR-compensated with the re-measured 777.5 GHz value; fit parameters are included in Table 1.

5 Conclusion and outlook

We have built and characterized a fiber Mach–Zehnder interferometer that measures the frequency-dependent reflection phase and amplitude of SiV nanophotonic cavities, and wrapped it in an automated pipeline that runs the entire measurement—probe-laser tuning, mode verification, power and lock calibration, the interleaved lock–probe cycles, FSR measurement and compensation, and the joint cavity fit—without manual intervention. The instrument’s noise budget is dominated by the per-cycle phase repeatability of 3.1° (circular standard deviation over 400 cycles), which averages down with cycle number and is already far below the fringe wander it replaces: unlocked, the interferometer drifts through full fringes on the second timescale. The arm imbalance is balanced by the delay stage to below half a millimeter of optical path ($\nu_{\text{FSR}} = 725.5 \text{ GHz}$) and the measured remainder is subtracted digitally, so the cavity’s own phase profile emerges essentially background-free. Applied to eleven scans of nine devices, the joint phase–amplitude fits separate over- from under-coupled cavities unambiguously—including the near-critical case where a magnitude-only measurement is maximally uninformative—and repeated measurements of the same devices on different days reproduce κ at the $\sim 15\%$ level, limited by the device–fiber interface rather than by the interferometer. The coupling-regime question, previously ambiguous from intensity data alone, is now a routine screening measurement.

Several upgrades would widen the instrument’s reach or remove its present bottlenecks:

- **A probe laser with a larger mode-hop-free tuning range.** Mode checking and re-tuning currently dominate the scan overhead near mode boundaries (Sections 3.3 and 3.4); a source that tunes continuously over the full scan window would remove most of the per-point tuning time and the multimode failure mode entirely.
- **Independent arm control with multiple Red Pitayas.** A second unit (or a multi-channel controller) would let the free-arm and delay-arm AOMs be driven and phase-controlled independently—e.g. locking on one arm while sweeping the other arbitrarily, or applying common-mode/differential decompositions of the actuation.
- **A more capable arbitrary-waveform generator** for the phase drive, with higher analog bandwidth, deeper waveform memory, and a wider external-modulation input range, would lift the $\pm v_{\text{sat}}$ clamp of Figure 10, allow faster and more complex sweep patterns, and reduce per-cycle dead time.
- **Replacing the tapered fiber.** The taper’s contact-mode coupling is the least reproducible element between sessions (Section 4.3); a permanently packaged coupling interface would stabilize both the extraction efficiency and the incoherent-background fraction g from run to run.
- **Polarization-maintaining fiber throughout.** The paddle controllers and in-fiber polarizers exist only because the arms are partly non-PM: polarization walk-off is what the 10 s visibility check guards against (Section 3.2). An end-to-end PM build would fix the interference contrast passively and eliminate a manual alignment step.
- **Acoustic and thermal isolation of the fiber paths.** The free-running spectrum (Figure 20) shows discrete acoustic lines and a broad 100–1000 Hz band on top of the thermal drift; enclosing the spools and dressing the patch fibers would shrink both the in-lock residual and the frozen-fringe drift during probe windows.
- **A faster lock.** Adding a proportional term and raising the unity-gain frequency (or moving the servo to a dedicated FPGA phase readout) would track the acoustic band actively rather than averaging over it, shortening the 0.3 s settling window that currently dominates the cycle time.

- **Automated FSR verification.** Periodically re-measuring the FSR and trimming the delay stage in closed loop would guard the sub-millimeter arm balance against slow fiber-length drift without operator attention.

A Automation flow diagrams

This appendix reproduces the control-software logic as flow diagrams, in the order a measurement session encounters them: system checks, probe-laser tuning and mode verification, the lock–probe measurement cycle, the FSR measurement, and the analysis pipeline.

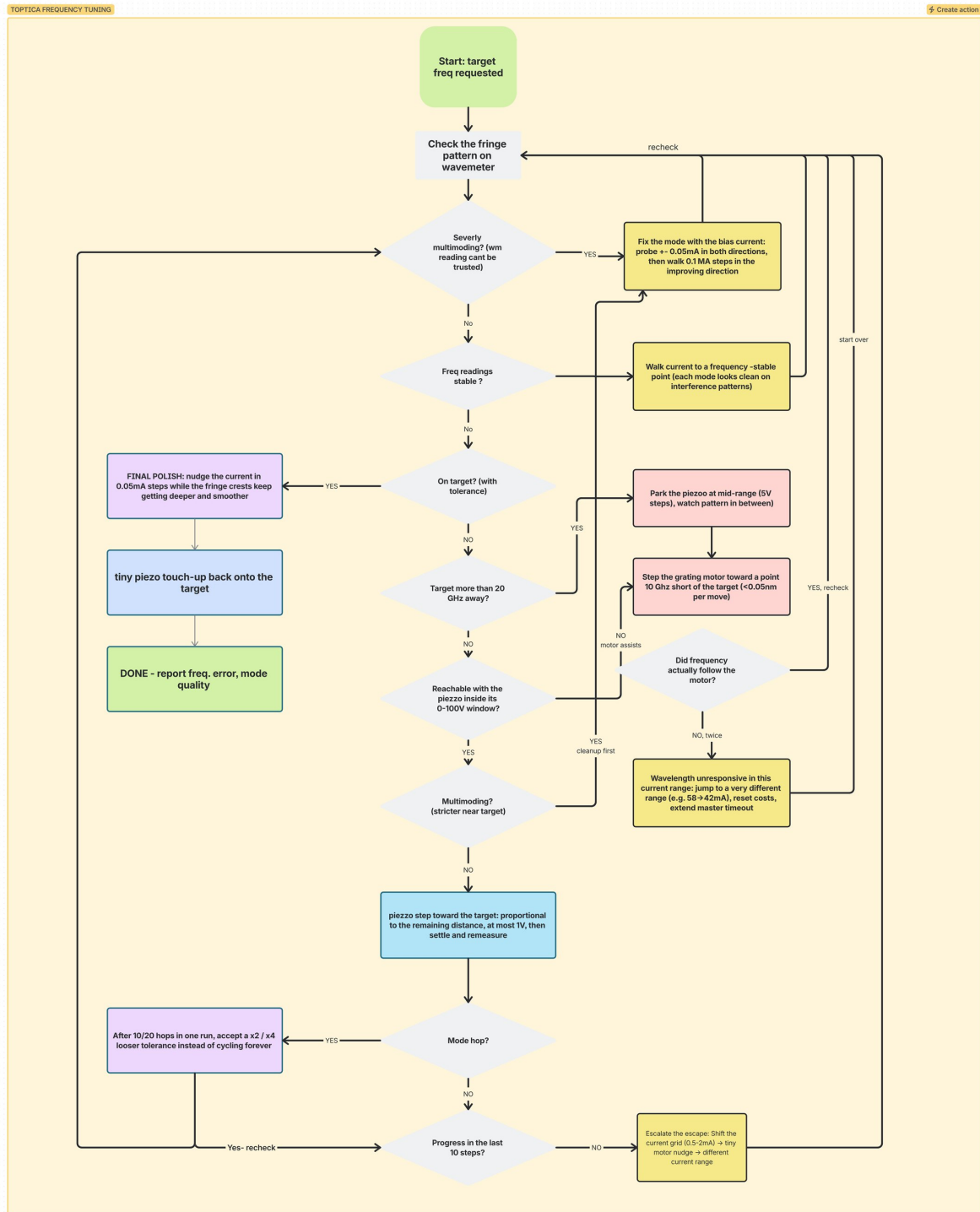


Figure 25: Probe-laser tuning algorithm (Section 3.4) as implemented: motor / piezo / current coordination, stability checks, and escape strategies.

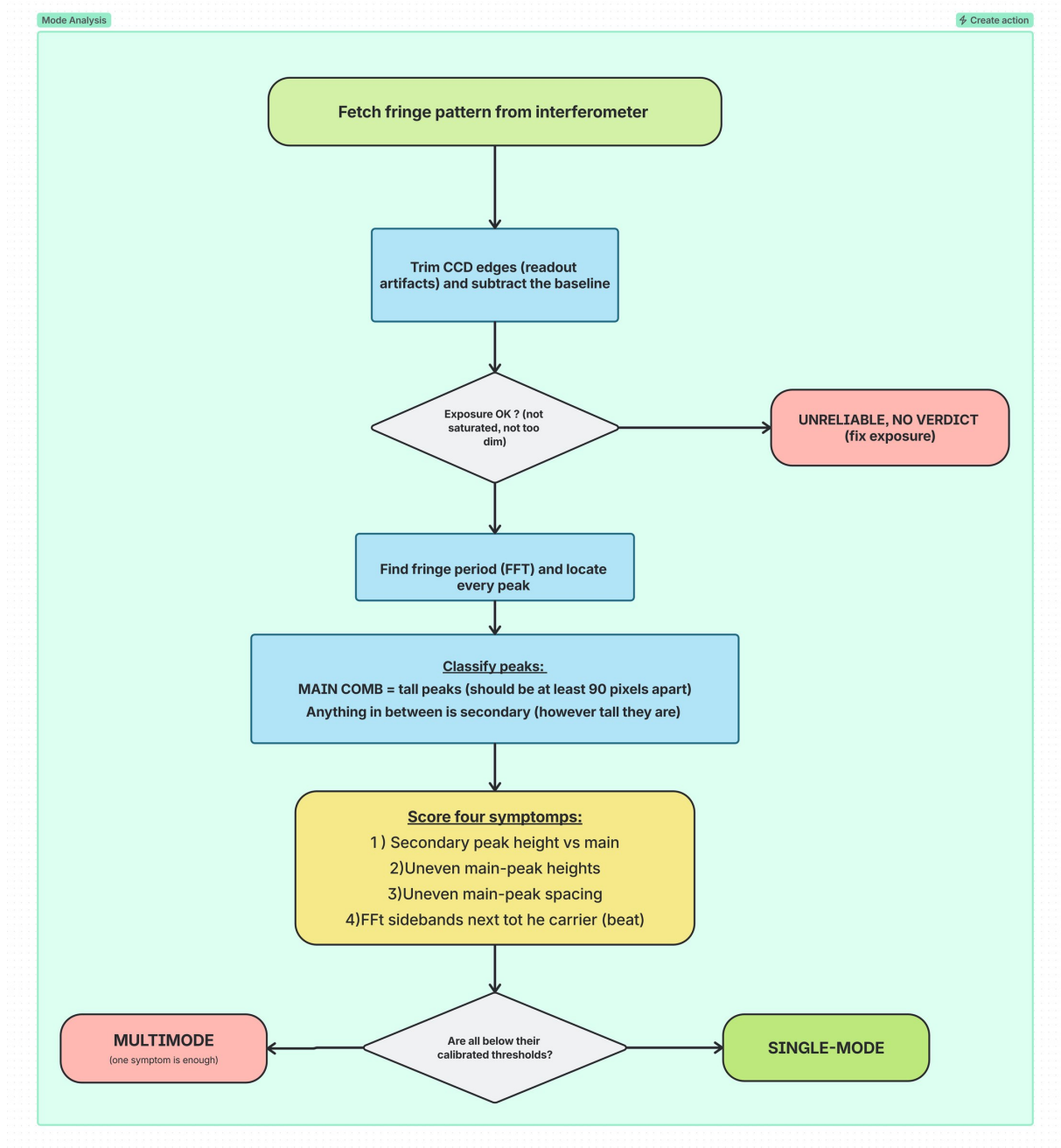


Figure 26: Mode-check classification (Section 3.3): fringe pattern acquisition, peak-comb classification, and the four scored multimode symptoms.

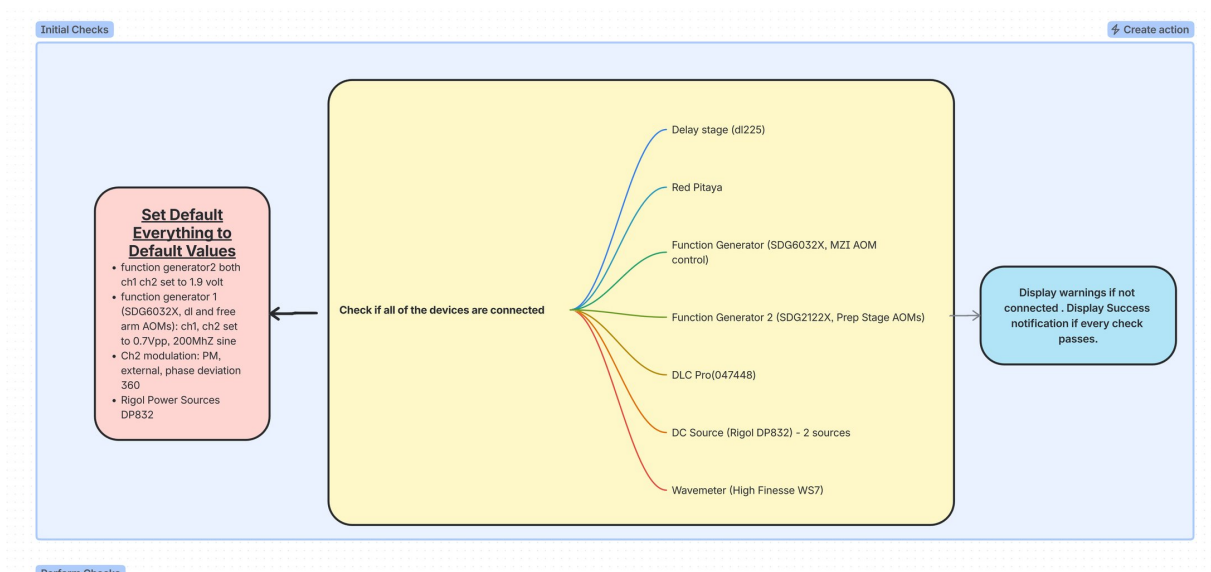


Figure 27: Initialization of device connections. Making sure connections are successful and devices have default values applied.

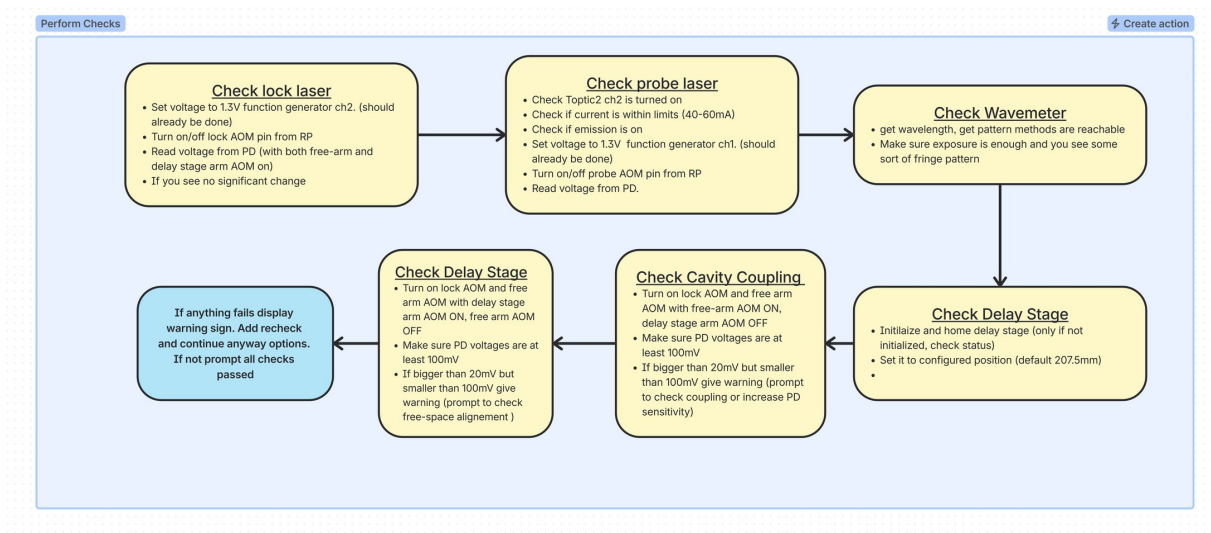


Figure 28: Pre-measurement system checks: device connection and default configuration, per-laser response checks, wavemeter and delay-stage verification, and the arm-coupling checks that gate the start of a scan.

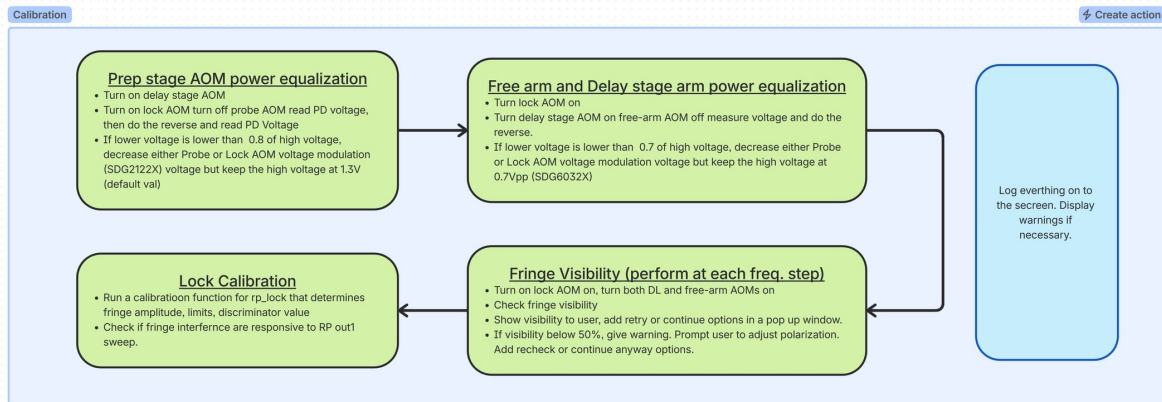


Figure 29: Calibration of AOMs, PI loop and visibility. Power equalization step works by changing AOM voltages and measuring power through PDs and making adjustment iteratively until required threshold is satisfied.

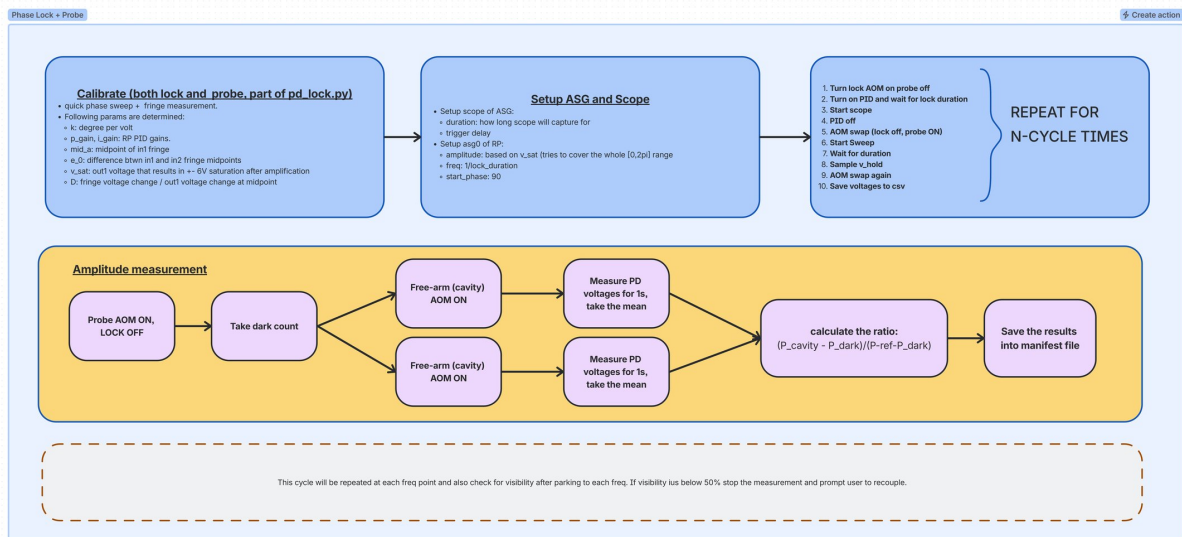


Figure 30: The lock-probe measurement engine (Section 3.1): calibration outputs, sweep and scope setup, the N -cycle lock-freeze-swap-sweep loop, and the arm-selective amplitude measurement appended at every frequency point.

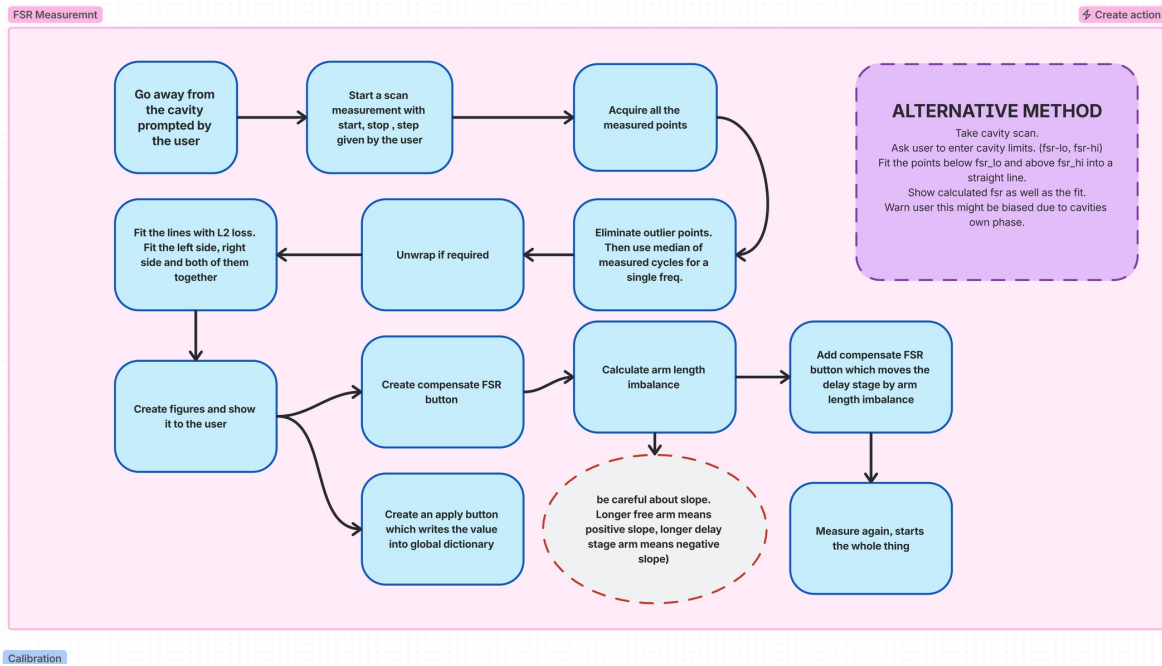


Figure 31: FSR measurement and compensation flow (Section 3.5): cavity-free scan, line fit, arm-imbalance calculation, and the delay-stage move with its double-pass conversion.

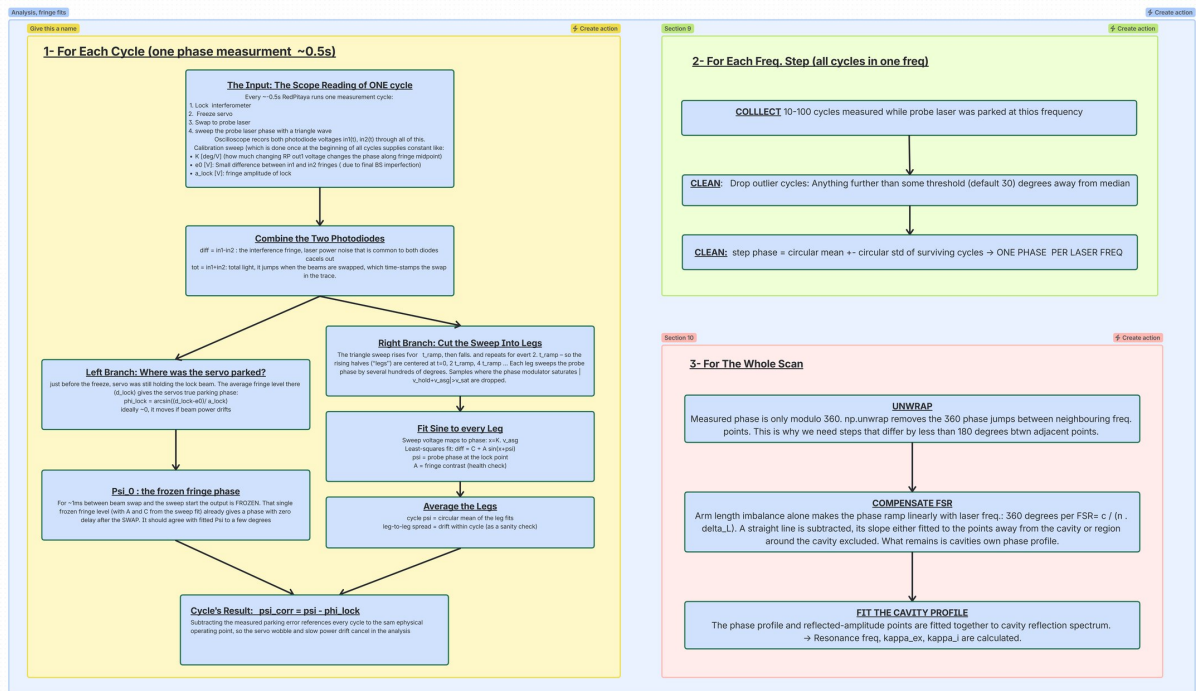


Figure 32: Analysis pipeline (Section 3.6): per-cycle reduction, per-frequency-step cleaning and averaging, and the whole-scan unwrap-compensate-fit sequence.

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